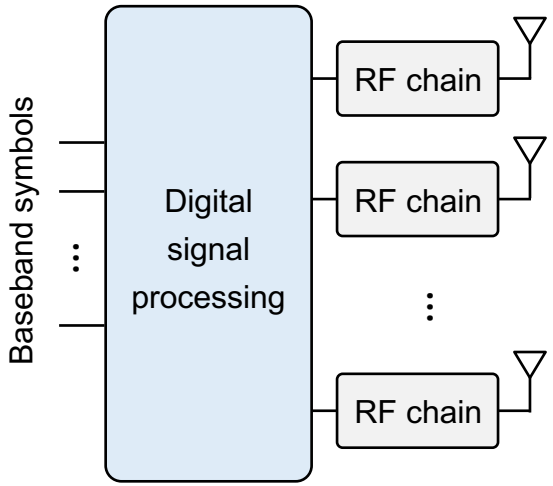
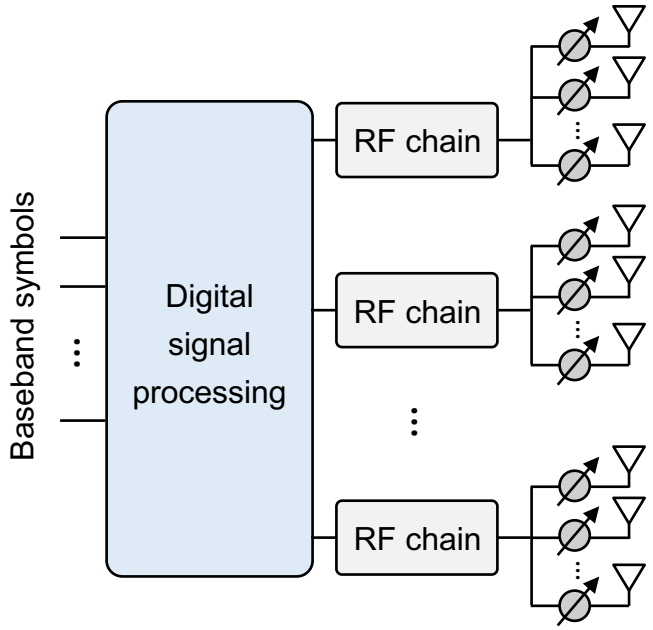
Outline

1. Background
2. Antenna Coding
3. Signal Processing for Reconfigurable Antennas
4. Reconfigurable Antennas for ISAC Applications
5. **Reconfigurable Antennas for Tri-Hybrid MIMO**
6. Prototyping and Experiments
7. Conclusions

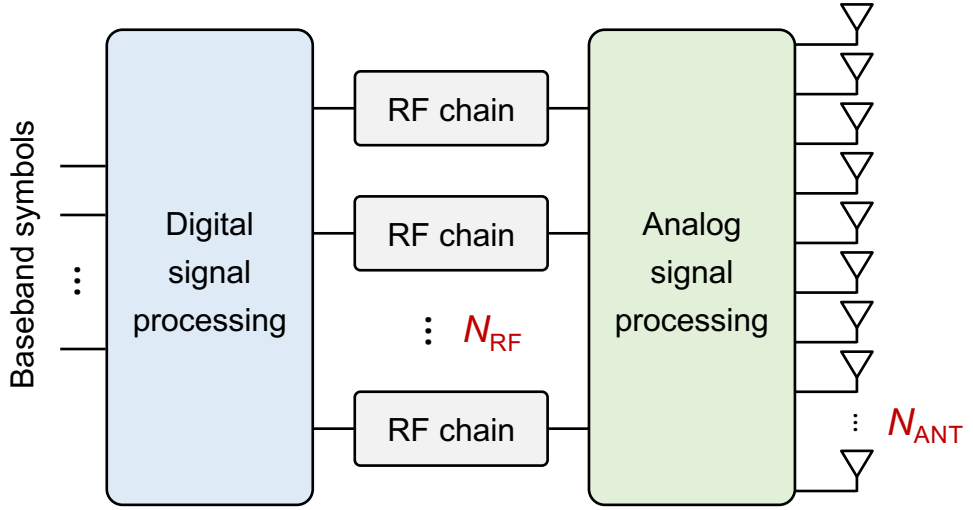
Evolution from Digital to Hybrid Digital-Analog MIMO



Conventional fully digital MIMO
(sub-6 GHz)



One-stream-per-subarray architecture
(mmWave or sub-THz)



General hybrid digital-analog MIMO
($N_{RF} < N_{ANT}$)

- ❑ Enjoy spatial diversity and multiplexing of large-scale antenna arrays
- ❑ Avoid scaling up energy consumption and hardware costs proportionally

»»

Move part of processing to the **analog domain**!

R. W. Heath, N. González-Prelcic, S. Rangan, W. Roh and A. M. Sayeed, "An Overview of Signal Processing Techniques for Millimeter Wave MIMO Systems," in IEEE Journal of Selected Topics in Signal Processing, vol. 10, no. 3, pp. 436-453, April 2016.

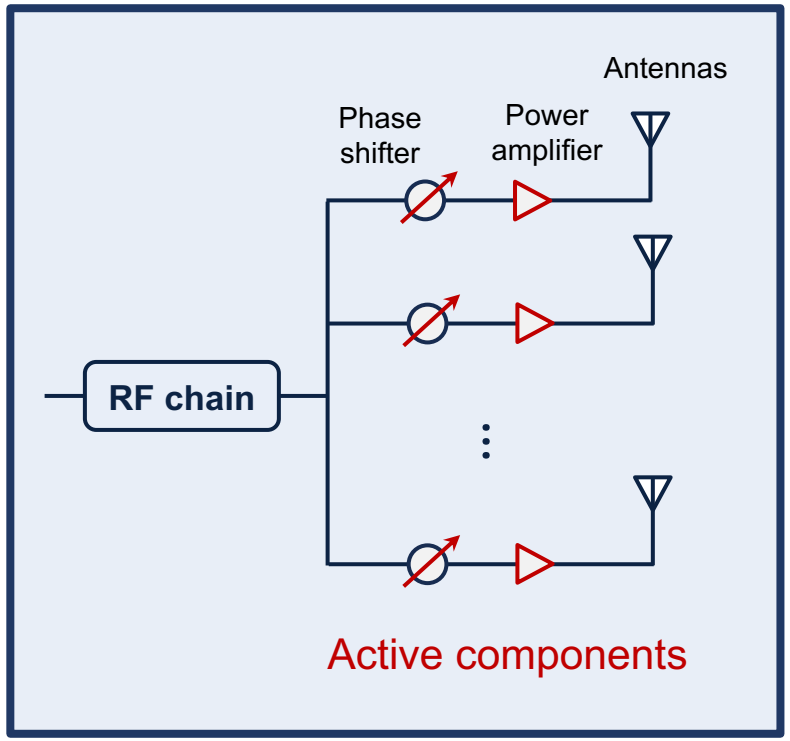
Energy-Efficiency Challenge in Massive MIMO

Deployment Scenarios For 6G RAN Evaluation in 3GPP TR 38.914 [1]

Carrier Frequency	System Bandwidth	BS Antenna Elements
700 MHz	Up to 60 MHz	Up to 64
2 GHz	Up to 200 MHz	Up to 288
4 GHz	Up to 300 MHz	Up to 576
7 GHz	Up to 400 MHz	Up to 2304
15 GHz	Up to 400 MHz	Up to 2304
30 GHz	Up to 1 GHz	Up to 4096

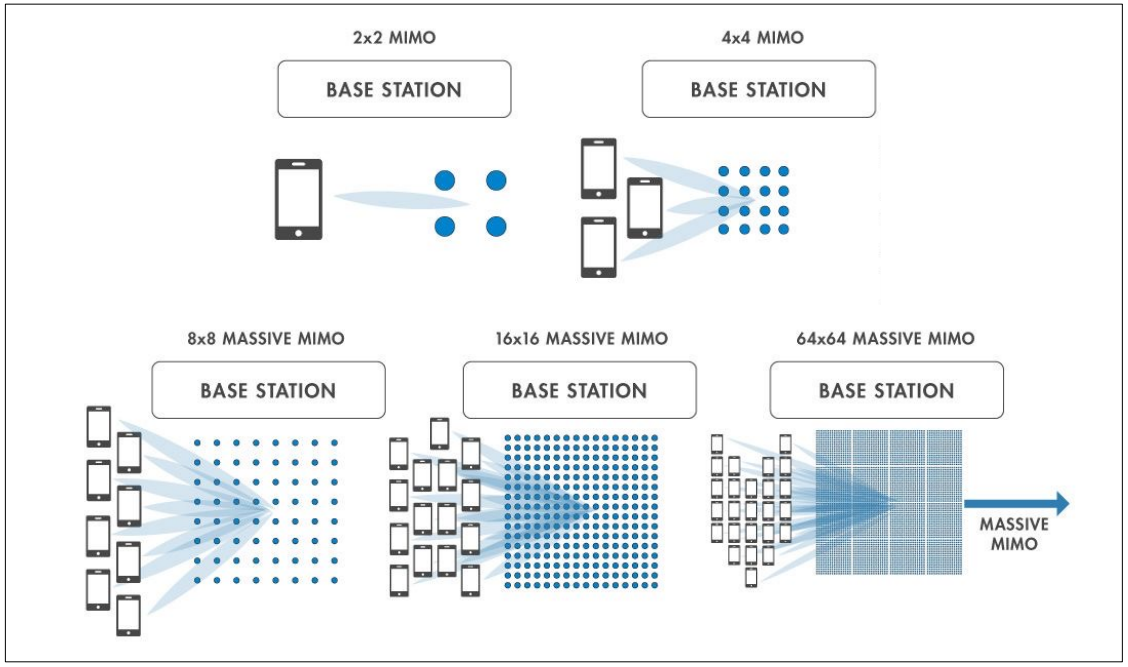
Energy consumption is becoming a major concern!

Analog processing

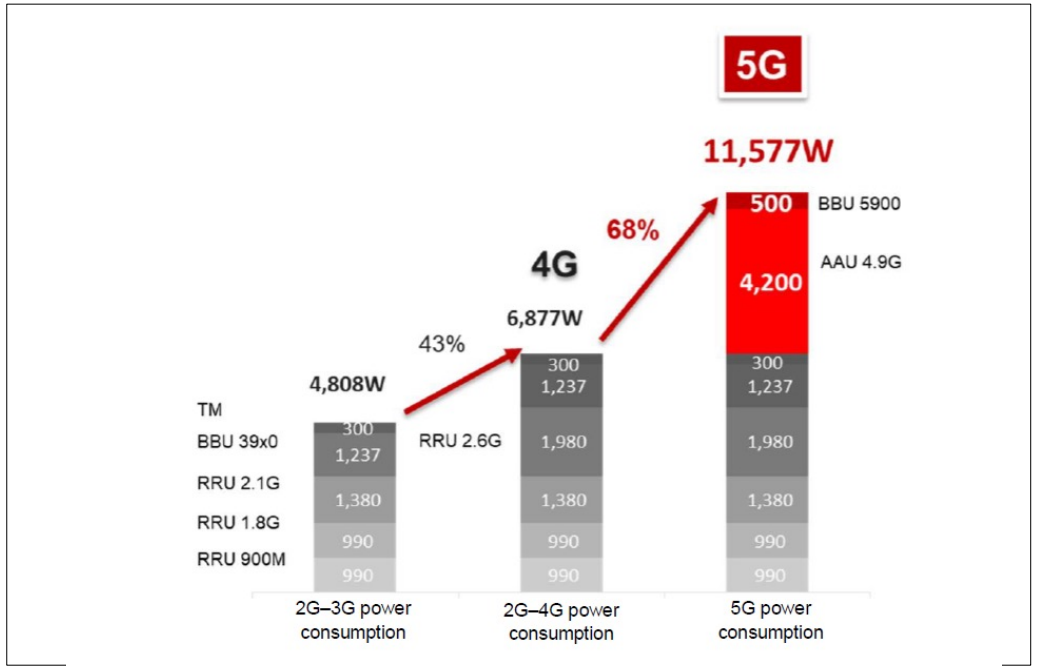


"Study on 6G Scenarios and Requirements," 3rd Generation Partnership Project (3GPP), Technical Specification Group Radio Access Network, 3GPP Technical Report TR 38.914 V0.2.0, Sep. 2025, (Release 20).

Energy-Efficiency Challenge in Massive MIMO



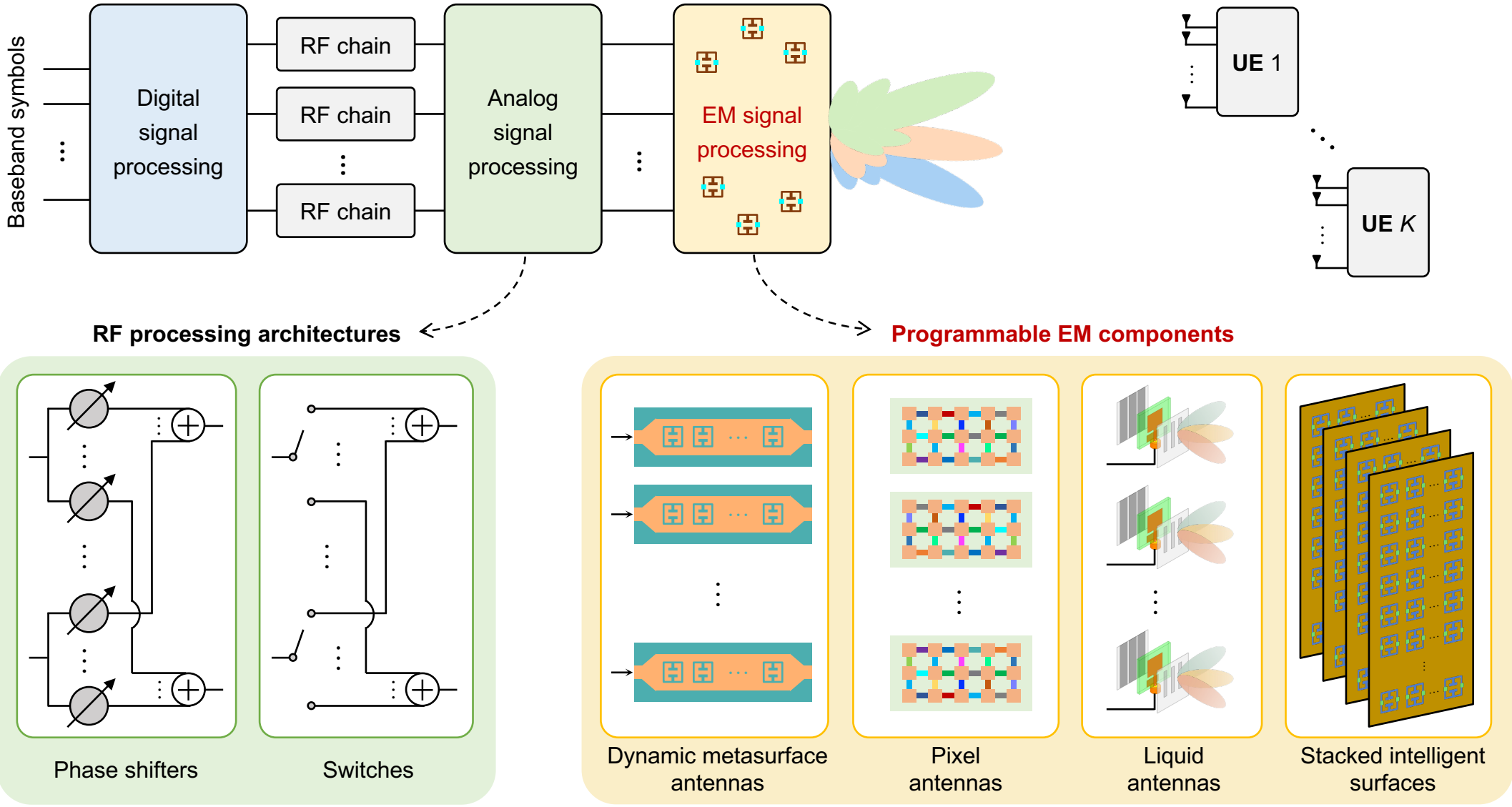
MIMO -> massive MIMO -> XL-MIMO -> ?



5G power consumption has increased by 68%. How about 6G?

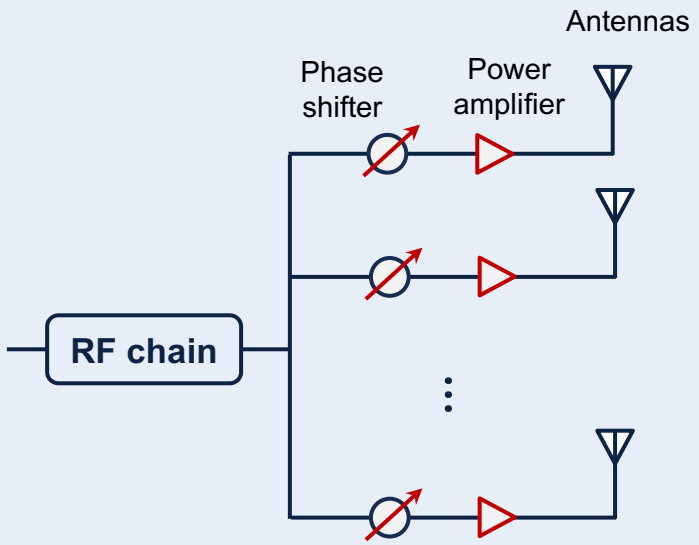
6G needs more energy-efficient solutions!

Tri-Hybrid MIMO: Move Part of Processing to EM Domain



EM Processing Is Cheaper

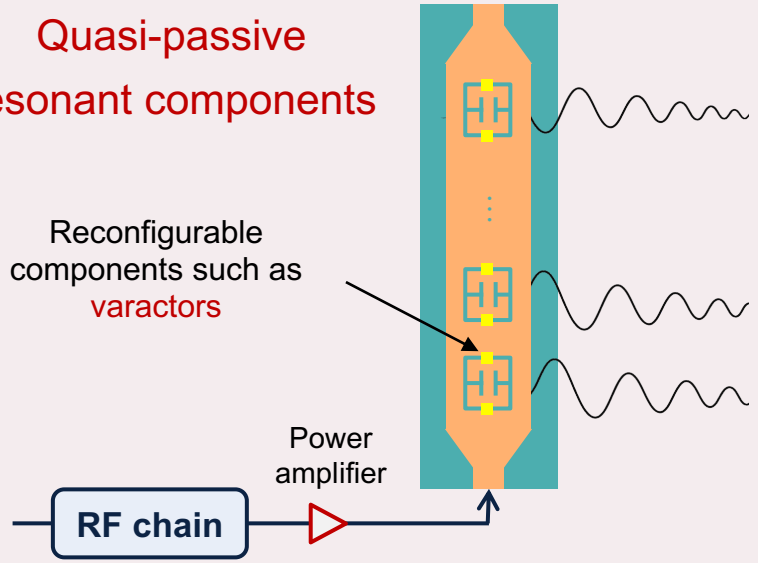
Analog processing



Active RF components

- Phase shifters realize accurate phase shifting control
- High power consumption (typically **tens to hundreds of mW** per phase shifter)

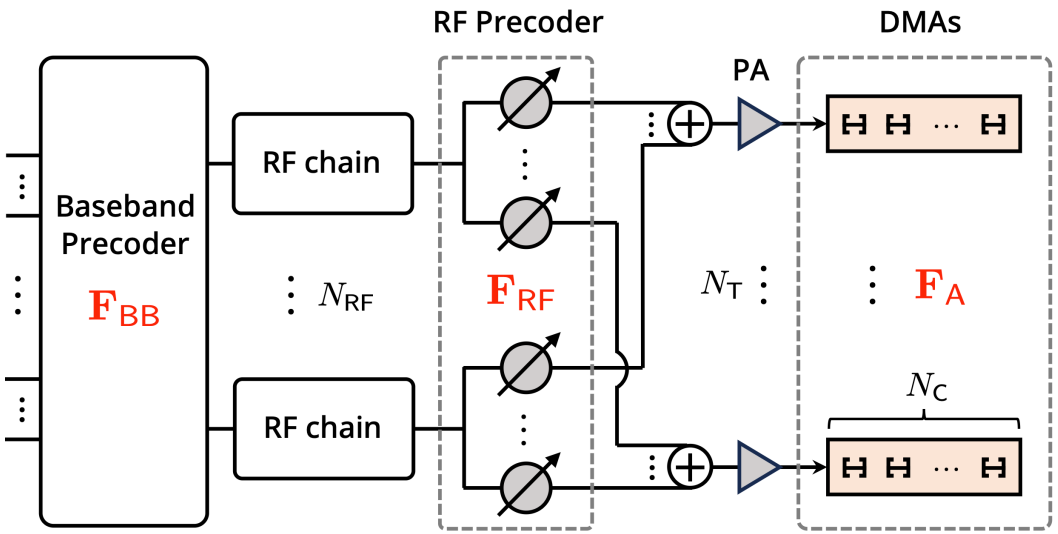
EM processing

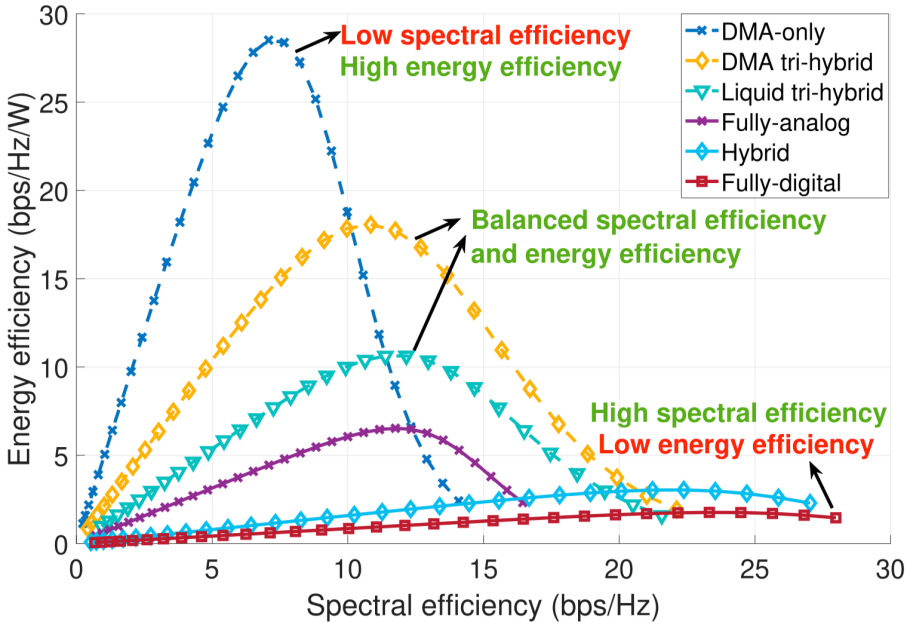


- Voltage-controlled variable capacitors
- Tune element resonant response
- Constrained beamforming but ultra-low power consumption (typically ***μ*W or lower** per varactor)

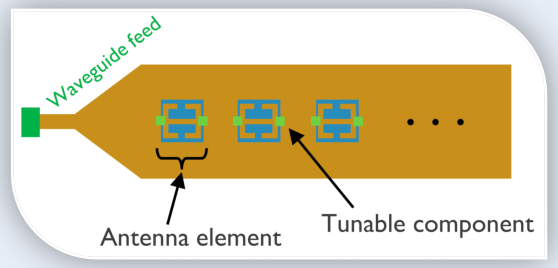
M. R. Castellanos, S. Yang, C. -B. Chae and R. W. Heath, "Embracing Reconfigurable Antennas in the Tri-Hybrid MIMO Architecture for 6G and Beyond," in IEEE Transactions on Communications, vol. 74, pp. 381-401, 2026.

EM Processing Is Cheaper





Dynamic metasurface antenna (DMA):



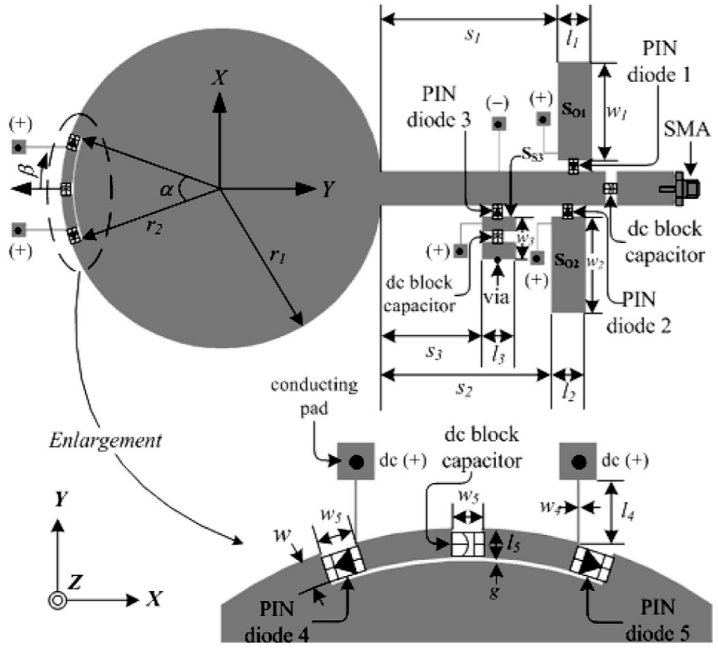
- ❑ Weaker beamforming capability
- ❑ But much lower power consumption!

Hybrid use of **phase shifters** and **EM tuners** achieves a decent balance between spectral efficiency and energy efficiency.

R. W. Heath, J. Carlson, N. V. Deshpande, M. R. Castellanos, M. Akrout and C. -B. Chae, "The Tri-Hybrid MIMO Architecture," in IEEE Wireless Communications, vol. 33, no. 1, pp. 199-206, Feb. 2026.

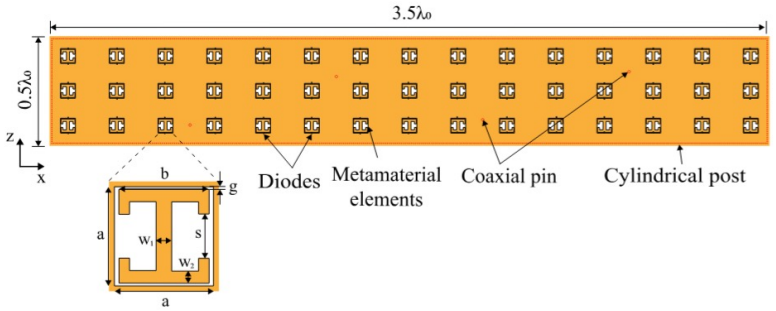
Various EM Processing Realizations

Electrically reconfigurable antenna: reconfigure antenna topology through RF switches or varactor diodes

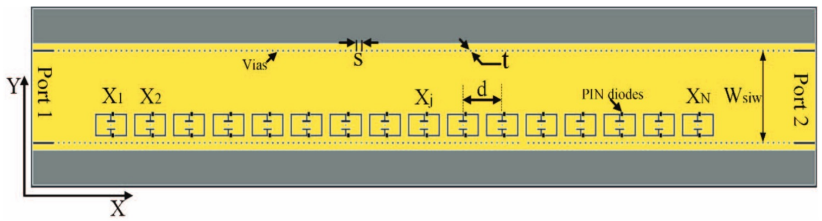


DOI: 10.1109/TAP.2008.916894

Traditional Reconfigurable Antenna

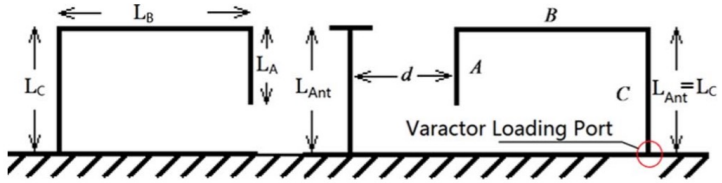


DOI: 10.1109/LWC.2020.2983374

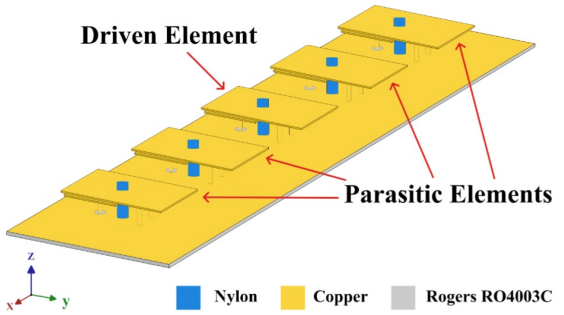


DOI: 10.1109/TAP.2020.3023581

Dynamic metasurface antennas (DMAs)



DOI: 10.1109/TAP.2011.2180300



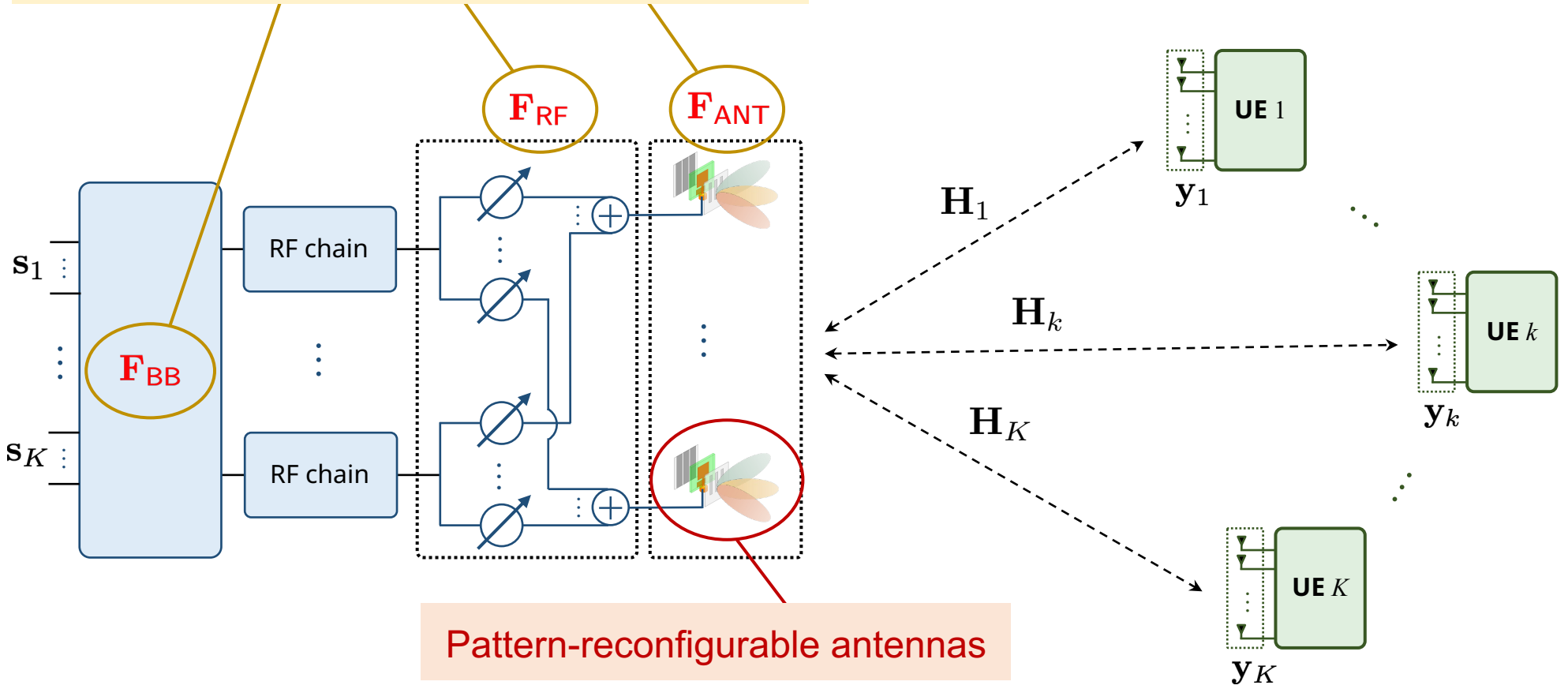
DOI: 10.1109/LAWP.2026.3712014

Electronically steerable parasitic array radiators (ESPARs)

Compact, reliable, and fast-switching to realize reconfigurability compared to mechanical solutions

Tri-Hybrid MIMO Precoding

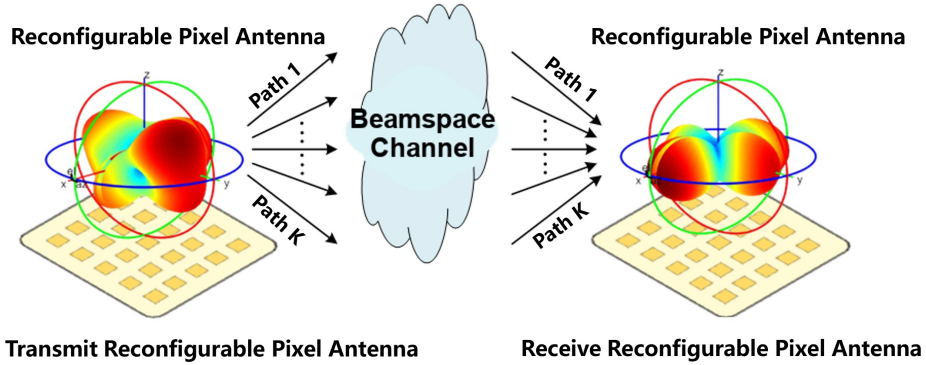
How to design these **tri-domain precoders?**



P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Model Pattern Reconfigurability

$$\mathbf{H}_k = \sqrt{\frac{NM_k}{L_k}} \sum_{\ell=1}^{L_k} \alpha_{k,\ell} G_{\mathsf{UE}}(\vartheta_{k,\ell}, \varphi_{k,\ell}) G_{\mathsf{BS}}(\theta_{k,\ell}, \phi_{k,\ell}) \mathbf{a}_{\mathsf{UE}}(\vartheta_{k,\ell}, \varphi_{k,\ell}) \mathbf{a}_{\mathsf{BS}}^{\mathsf{H}}(\theta_{k,\ell}, \phi_{k,\ell})$$



Finite Pattern Selection

- Select a pattern from S candidates
- Align with how practical hardware works
- For the *n*-th antenna:

$$G_{(n)}^{\mathsf{BS}}(\theta,\phi) = \mathbf{g}^{\mathsf{T}}(\theta,\phi)\mathbf{b}_{(n)}$$

S candidate patterns

2^Q patterns in pixel antennas

One-hot vector: $\mathbf{b}_{(n)} \in \{\mathbf{b} \in \{0,1\}^S : \|\mathbf{b}\|_0 = 1\}, \forall n$

- Define the **selection** precoder as:

$$\mathbf{F}_{\mathsf{sel}} = \mathrm{blkdiag}\{\mathbf{b}_{(1)}, \mathbf{b}_{(2)}, \ldots, \mathbf{b}_{(N)}\}$$

Arbitrary Pattern Generation

- Assume we can generate any patterns
- Serve as a performance upper bound
- Use **spherical harmonics expression**:

$$G_{(n)}^{\mathsf{BS}}(\theta,\phi) \approx \sum_{u=0}^{U} \sum_{q=-u}^{u} c_{uq}^{(n)} Y_u^q(\theta,\phi) = \boldsymbol{\gamma}^{\mathsf{T}}(\theta,\phi)\mathbf{c}_{(n)}$$

Harmonic basis

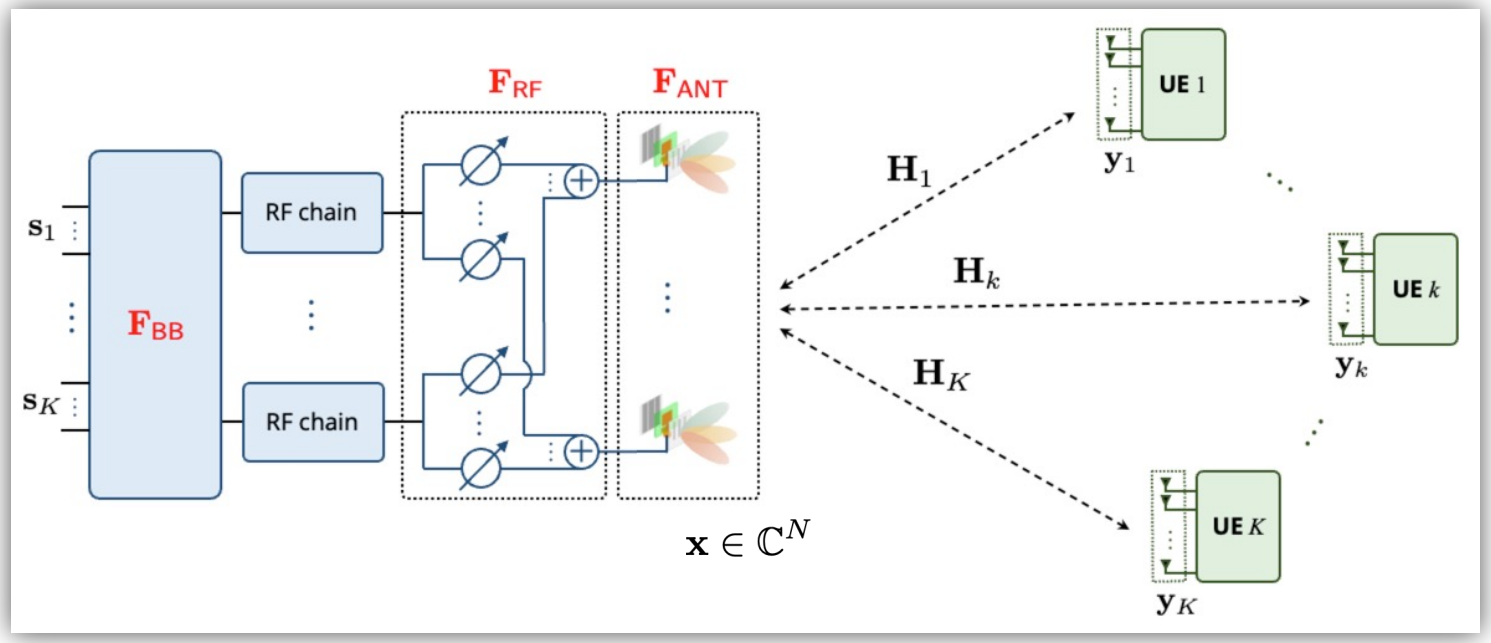
Coefficient vector

- Define the **coefficient** precoder as:

$$\mathbf{F}_{\mathsf{cof}} = \mathrm{blkdiag}\{\mathbf{c}_{(1)}, \mathbf{c}_{(2)}, \ldots, \mathbf{c}_{(N)}\}$$

P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Formulate the Precoder Optimization Problem



Transmitted baseband symbol:

$$\mathbf{s} = [\mathbf{s}_1^{\mathsf{T}}, \mathbf{s}_2^{\mathsf{T}}, \ldots, \mathbf{s}_K^{\mathsf{T}}]^{\mathsf{T}}$$

Transmitted RF signal:

$$\mathbf{x} = \mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB}}\mathbf{s}$$

$$= \mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}} \left[\mathbf{F}_{\text{BB},1}, \ldots, \mathbf{F}_{\text{BB},K}\right] \begin{bmatrix} \mathbf{s}_1 \\ \vdots \\ \mathbf{s}_K \end{bmatrix}$$

$\mathbf{F}_{\text{ANT}} \in \{\mathbf{F}_{\text{sel}}, \mathbf{F}_{\text{cof}}\}$

Received signal at user k: $\mathbf{y}_k = \mathbf{H}_k\mathbf{x} = \mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB}}\mathbf{s}$

$$= \mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},k}\mathbf{s}_k + \sum_{i \neq k} \mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},i}\mathbf{s}_i + \mathbf{n}_k$$

Desired signal for user *k* | Interference | Noise

P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Formulate the Precoder Optimization Problem

Received signal at user k: $\mathbf{y}_k = \mathbf{H}_k\mathbf{x} = \mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB}}\mathbf{s}$

$$= \mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},k}\mathbf{s}_k + \sum_{i\neq k}\mathbf{H}_k\mathbf{F}_{\text{ANT}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},i}\mathbf{s}_i + \mathbf{n}_k$$

Desired signal for user k | Interference | Noise

Let $\mathbf{F}_{\text{ANT}} = \mathbf{F}_{\text{sel}}$, we can formulate the **sum-rate maximization problem**:

$$\max_{\mathbf{F}_{\text{sel}},\mathbf{F}_{\text{RF}},\mathbf{F}_{\text{BB}}} \sum_{k=1}^{K}\beta_k \log_2\det\left(\mathbf{I}_{M_k} + \mathbf{H}_k^{\text{sel}}\mathbf{F}_{\text{sel}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},k}\mathbf{F}_{\text{BB},k}^{\text{H}}\mathbf{F}_{\text{RF}}^{\text{H}}\mathbf{F}_{\text{sel}}^{\text{H}}\left(\mathbf{H}_k^{\text{sel}}\right)^{\text{H}}\right.$$

Power of desired signal

$$\left.\times\left(\sum_{i\neq k}\mathbf{H}_k^{\text{sel}}\mathbf{F}_{\text{sel}}\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB},i}\mathbf{F}_{\text{BB},i}^{\text{H}}\mathbf{F}_{\text{RF}}^{\text{H}}\mathbf{F}_{\text{sel}}^{\text{H}}\left(\mathbf{H}_k^{\text{sel}}\right)^{\text{H}} + \sigma_k^2\mathbf{I}_{M_k}\right)^{-1}\right)$$

Power of interference & noise

$$\text{s.t.} \quad \left[\mathbf{F}_{\text{RF}}\mathbf{F}_{\text{BB}}\mathbf{F}_{\text{BB}}^{\text{H}}\mathbf{F}_{\text{RF}}^{\text{H}}\right]_{n,n} \leq P_n,\ \forall n,$$

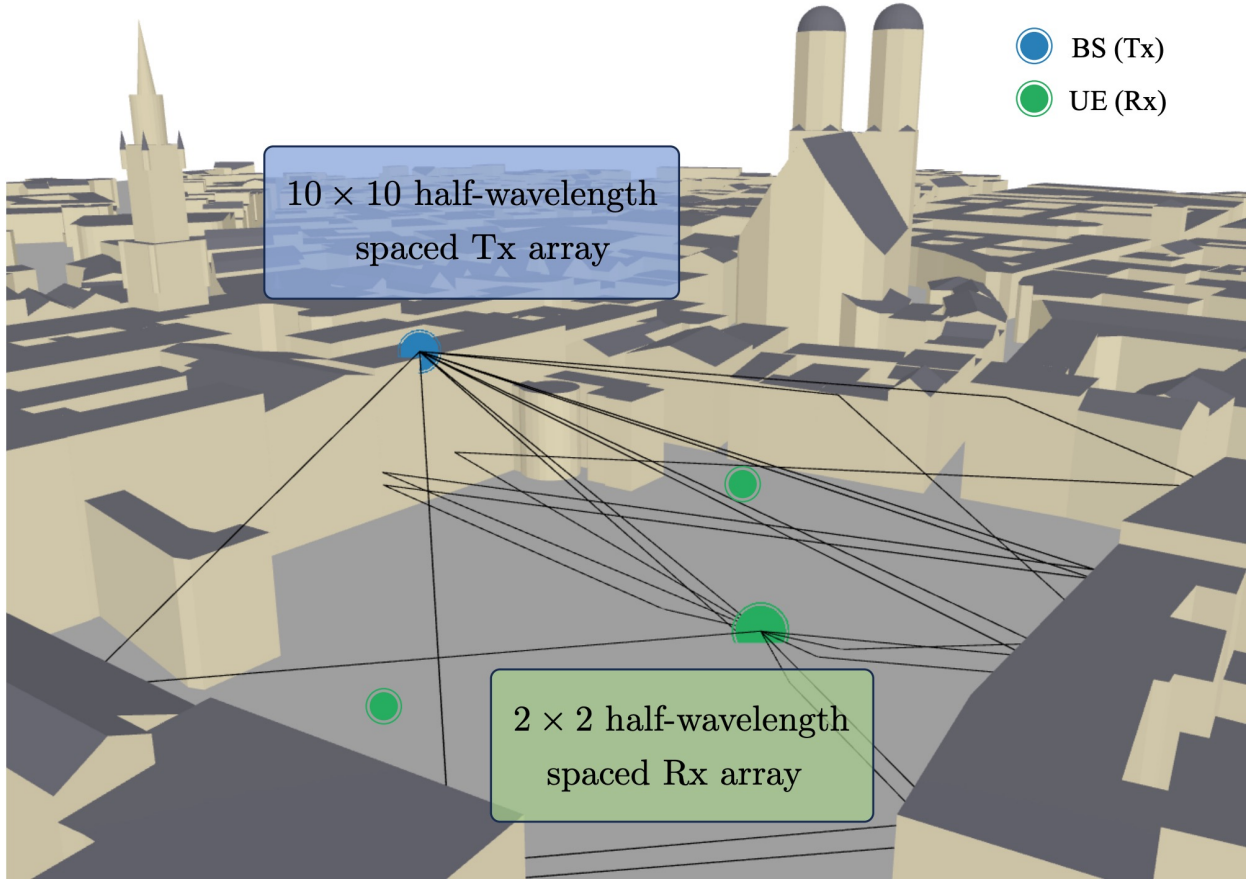
$$\mathbf{F}_{\text{sel}} = \text{blkdiag}\left\{\mathbf{b}_{(1)},\mathbf{b}_{(2)},\ldots,\mathbf{b}_{(N)}\right\},$$

$$\mathbf{b}_{(n)} \in \left\{\mathbf{b}\in\{0,1\}^S \mid \|\mathbf{b}\|_0 = 1\right\},\ \forall n.$$

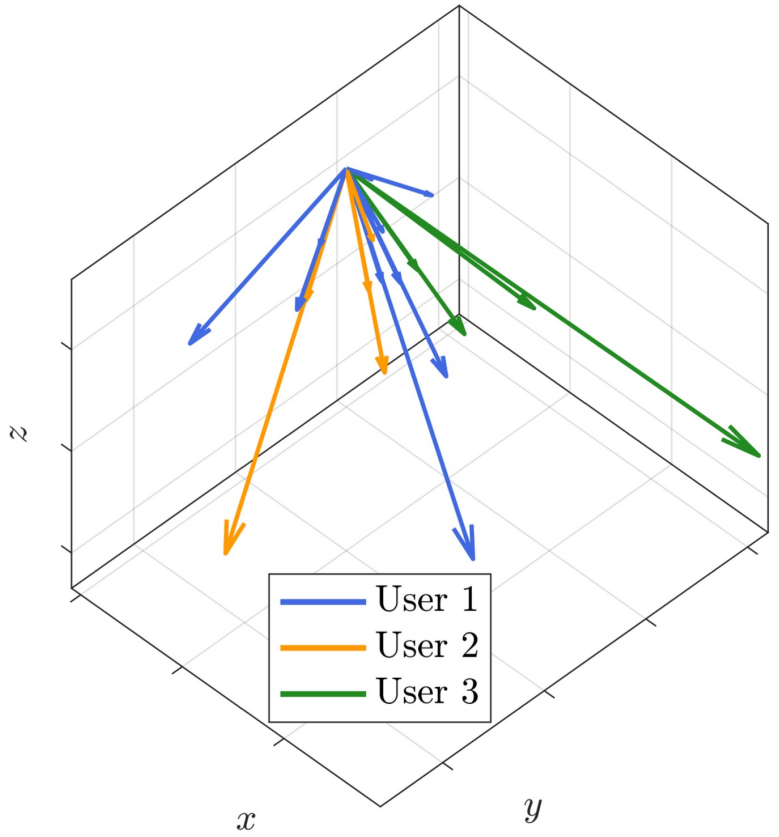
P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Performance Evaluation

Ray-tracing channel simulation via Sionna RT



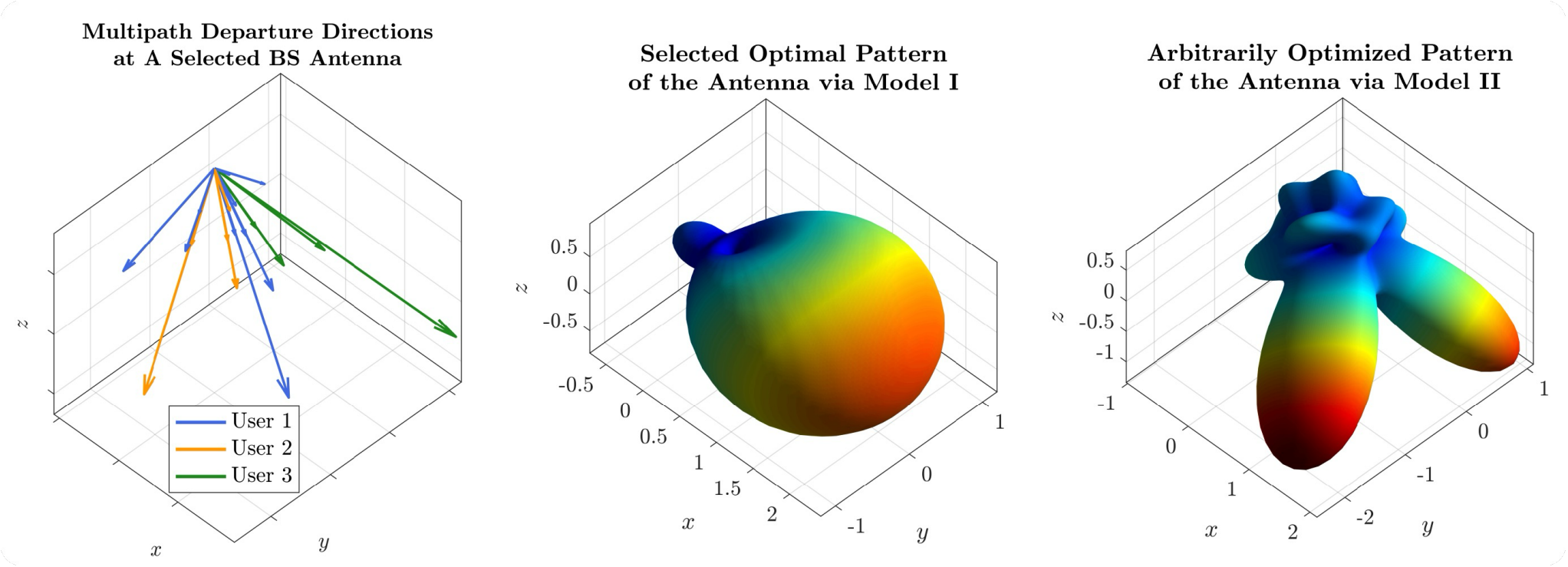
Multipath departure directions



P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Performance Evaluation

Visualization of the optimized radiation pattern of a single antenna

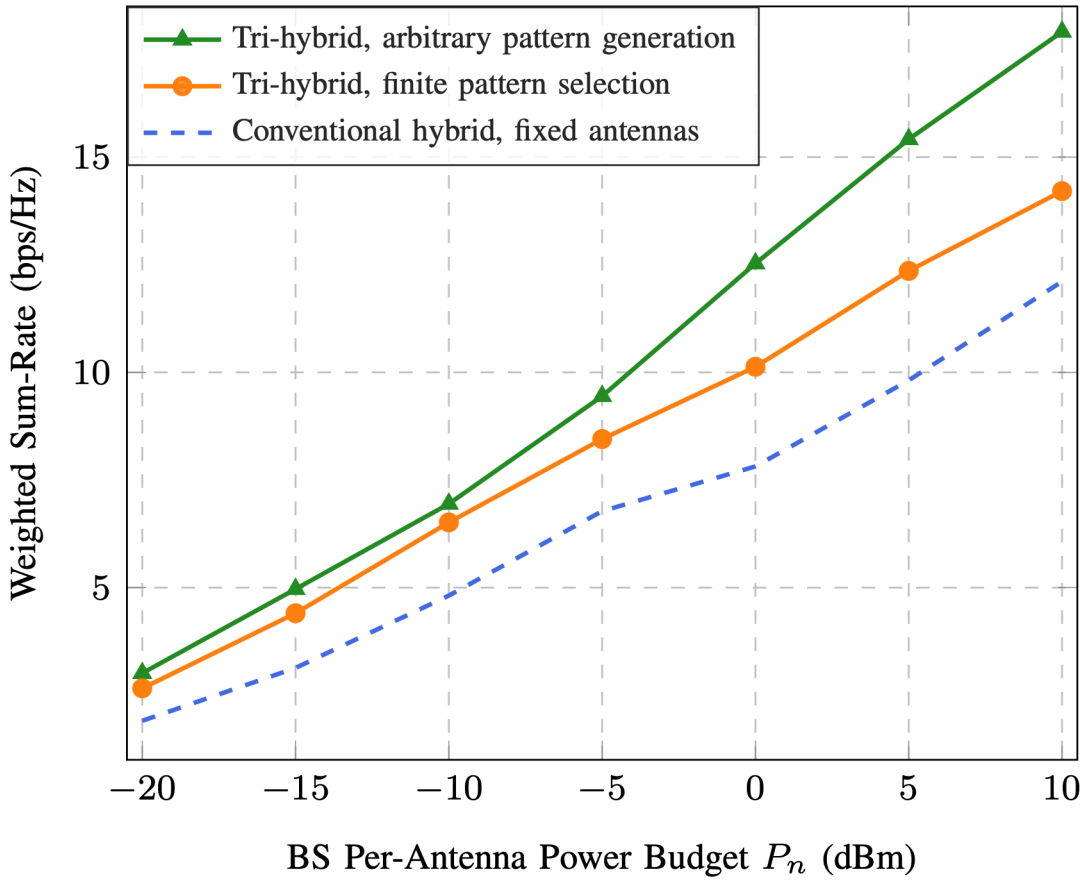


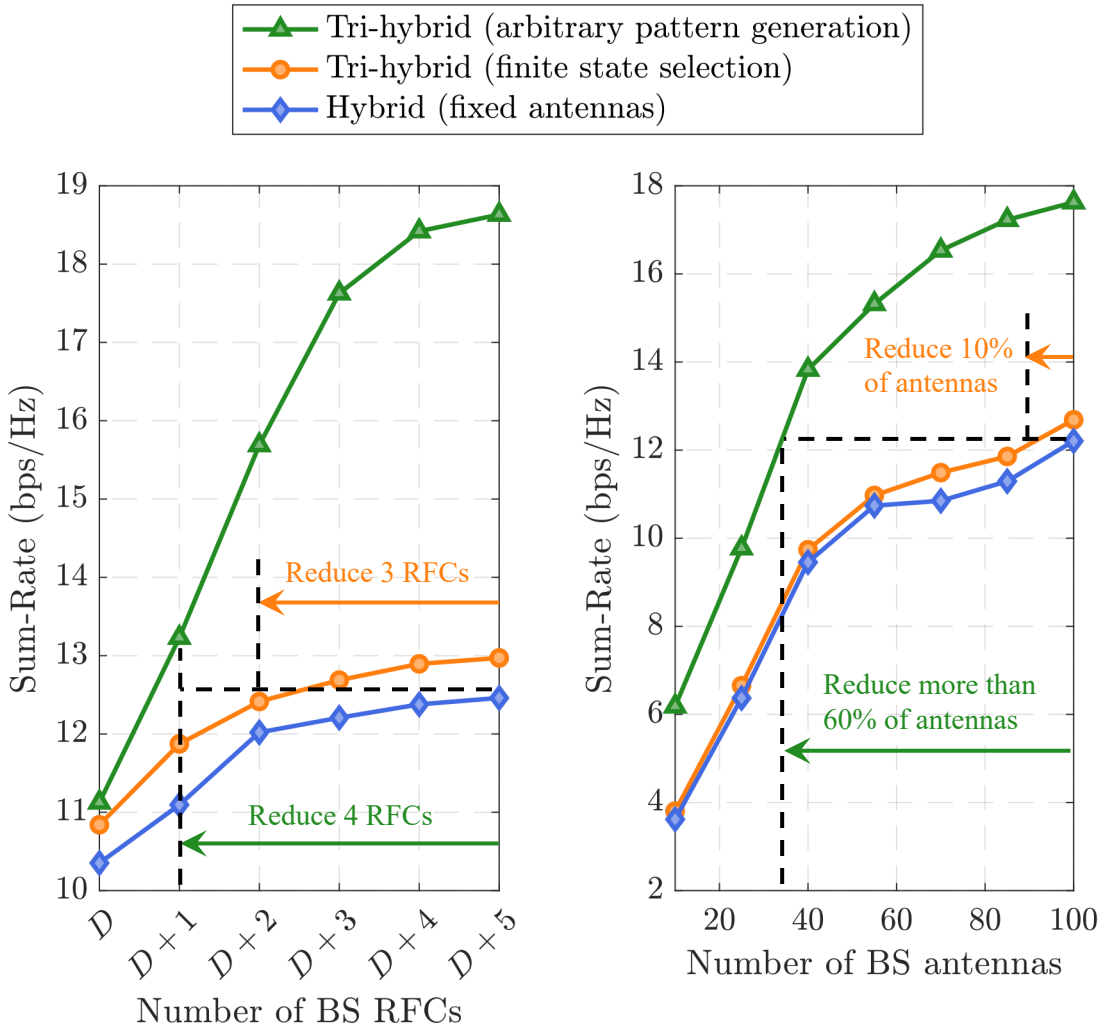
Finite pattern selection
(based on a fluid RA prototype)

Arbitrary pattern generation

P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Performance Evaluation





P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Lessons Learned

Key Insights

- Tri-hybrid MIMO holds significant potential for multiuser communications.
- Tri-hybrid MIMO is promising in enhancing hardware efficiency.

Unsolved Challenges

- The formulated optimization problem is very complex.
- Using iterative algorithm to solve it is very time-consuming and thus impractical.

P. Zheng, Y. Zhang, T. Y. Al-Naffouri, M. J. Hossain and A. Chaaban, "Tri-Hybrid Multi-User Precoding Using Pattern-Reconfigurable Antennas: Fundamental Models and Practical Algorithms," in IEEE Transactions on Communications, 2026, early access.

Iterative Alg. for Solving the Tri-Hybrid Precoding Problem

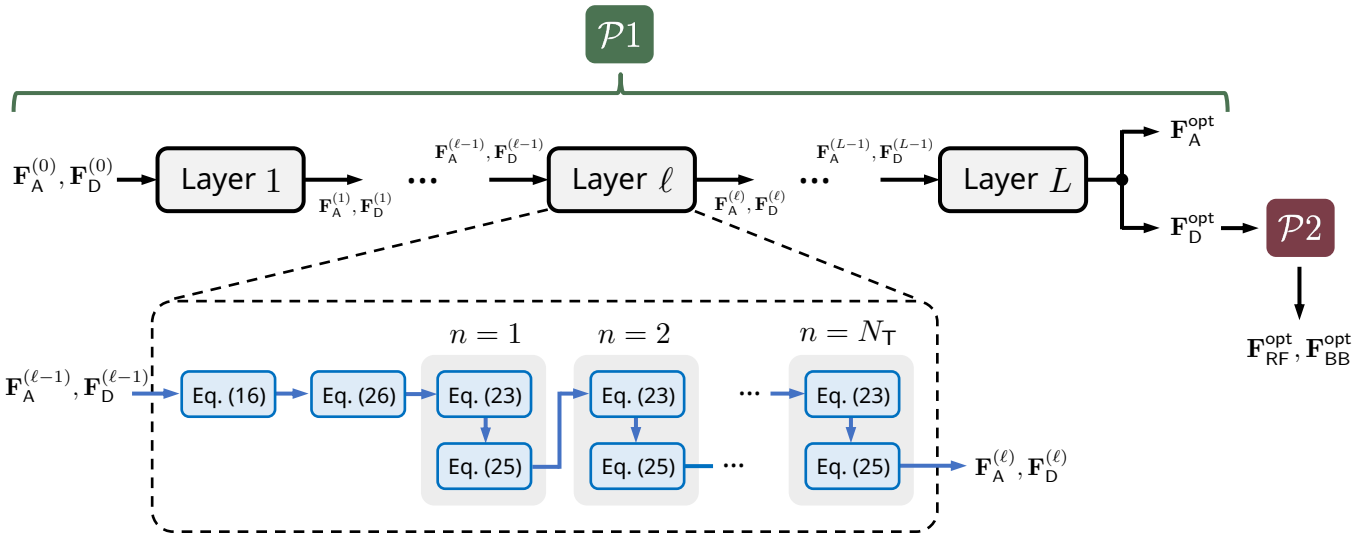
$$\max_{\mathbf{F}_{\mathsf{sel}},\mathbf{F}_{\mathsf{RF}},\mathbf{F}_{\mathsf{BB}}} \sum_{k=1}^{K} \beta_k \log_2 \det \Big(\mathbf{I}_{M_k} + \mathbf{H}_k^{\mathsf{sel}} \mathbf{F}_{\mathsf{sel}} \mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB},k} \mathbf{F}_{\mathsf{BB},k}^{\mathsf{H}} \mathbf{F}_{\mathsf{RF}}^{\mathsf{H}} \mathbf{F}_{\mathsf{sel}}^{\mathsf{H}} \big(\mathbf{H}_k^{\mathsf{sel}}\big)^{\mathsf{H}} \times \Big(\sum_{i\neq k} \mathbf{H}_k^{\mathsf{sel}} \mathbf{F}_{\mathsf{sel}} \mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB},i} \mathbf{F}_{\mathsf{BB},i}^{\mathsf{H}} \mathbf{F}_{\mathsf{RF}}^{\mathsf{H}} \mathbf{F}_{\mathsf{sel}}^{\mathsf{H}} \big(\mathbf{H}_k^{\mathsf{sel}}\big)^{\mathsf{H}} + \sigma_k^2 \mathbf{I}_{M_k} \Big)^{-1} \Big)$$

$$\text{s.t.} \quad \big[\mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB}} \mathbf{F}_{\mathsf{BB}}^{\mathsf{H}} \mathbf{F}_{\mathsf{RF}}^{\mathsf{H}}\big]_{n,n} \le P_n, \ \forall n,$$

$$\mathbf{F}_{\mathsf{sel}} = \mathrm{blkdiag}\big\{\mathbf{b}_{(1)}, \mathbf{b}_{(2)}, \ldots, \mathbf{b}_{(N)}\big\},$$

$$\mathbf{b}_{(n)} \in \big\{\mathbf{b} \in \{0,1\}^S \mid \|\mathbf{b}\|_0 = 1\big\}, \ \forall n.$$

$$\mathbf{F}_{\mathsf{D}} \triangleq \mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB}}$$



$\mathcal{P}1$

$$\max_{\mathbf{F}_{\mathsf{A}},\mathbf{F}_{\mathsf{D}}} \sum_{k=1}^{K} \beta_k \log_2 \det \Big(\mathbf{I} + \mathbf{H}_k \mathbf{F}_{\mathsf{A}} \mathbf{F}_{\mathsf{D},k} \mathbf{F}_{\mathsf{D},k}^{\mathsf{H}} \mathbf{F}_{\mathsf{A}}^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} \times \big(\sum_{i\neq k} \mathbf{H}_k \mathbf{F}_{\mathsf{A}} \mathbf{F}_{\mathsf{D},i} \mathbf{F}_{\mathsf{D},i}^{\mathsf{H}} \mathbf{F}_{\mathsf{A}}^{\mathsf{H}} \mathbf{H}_k^{\mathsf{H}} + \sigma_k^2 \mathbf{I} \big)^{-1} \Big)$$

$$\text{s.t.} \quad [\mathbf{F}_{\mathsf{D}} \mathbf{F}_{\mathsf{D}}^{\mathsf{H}}]_{n,n} \le P_n, \ \forall n,$$

$$\mathbf{F}_{\mathsf{A}} = \mathrm{blkdiag}\{\mathbf{b}_{(1)}, \mathbf{b}_{(2)}, \ldots, \mathbf{b}_{(N)},\}$$

$$\mathbf{b}_{(n)} \in \big\{\mathbf{b} \in \{0,1\}^S \mid \|\mathbf{b}\|_0 = 1\big\}, \ \forall n.$$

$\mathcal{P}2$

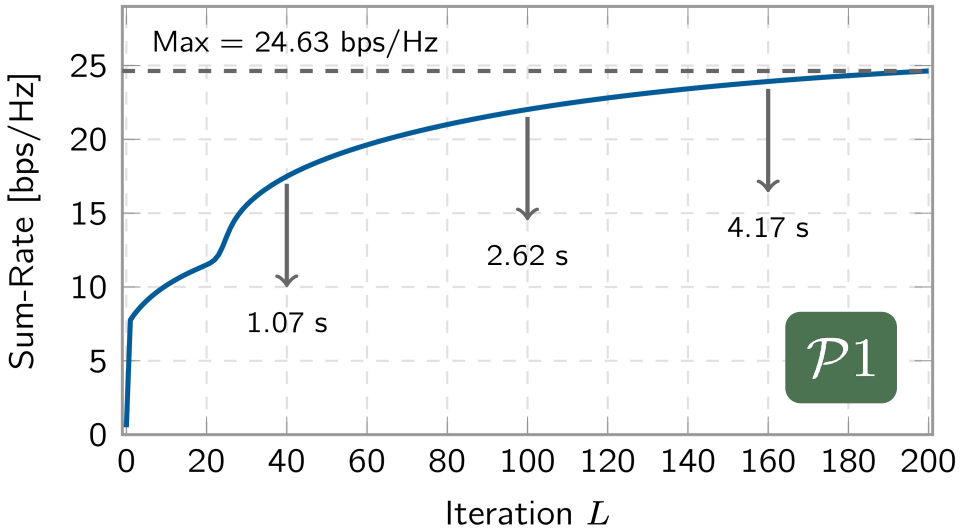
$$\min_{\mathbf{F}_{\mathsf{RF}},\mathbf{F}_{\mathsf{BB}}} \ \|\mathbf{F}_{\mathsf{D}}^{\mathsf{opt}} - \mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB}}\|_{\mathsf{F}}^2$$

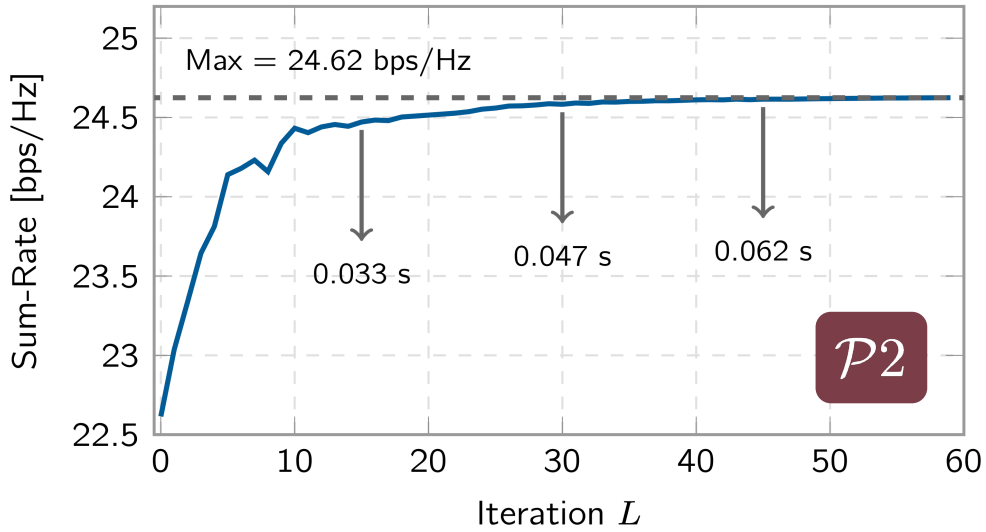
$$\text{s.t.} \quad |[\mathbf{F}_{\mathsf{RF}}]_{i,j}|^2 = 1/N_{\mathsf{T}}, \ \forall i,j,$$

$$[\mathbf{F}_{\mathsf{RF}} \mathbf{F}_{\mathsf{BB}} \mathbf{F}_{\mathsf{BB}}^{\mathsf{H}} \mathbf{F}_{\mathsf{RF}}^{\mathsf{H}}]_{n,n} \le P_n, \ \forall n.$$

P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

Iterative Alg. for Solving the Tri-Hybrid Precoding Problem

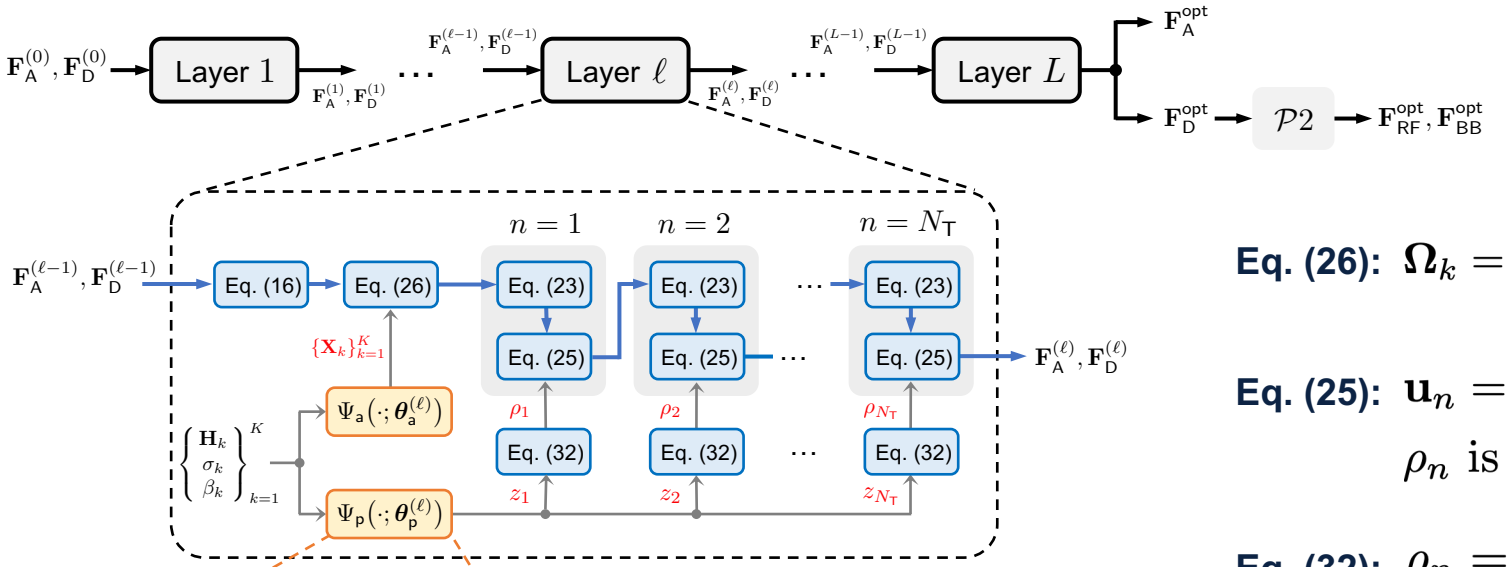




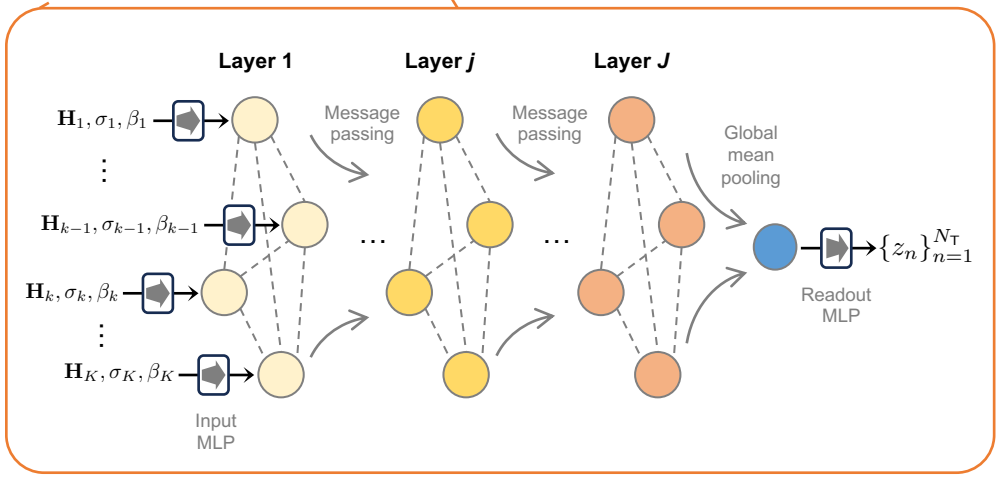
- P1 is the main bottleneck.
- The algorithm is too slow to be useful.

P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

Deep Algorithm Unfolding



Graph Neural Network



An acceleration term

Eq. (26): $\boldsymbol{\Omega}_k = \left(\mathbf{I} - \boldsymbol{\Gamma}_k^{\mathsf{H}}\mathbf{H}_k\mathbf{F}_{\mathsf{A}}\mathbf{F}_{\mathsf{D},k}\right)^{-1} + \mathbf{X}_k + \mathbf{X}_k{}^{\mathsf{H}},\ \forall k$

Eq. (25): $\mathbf{u}_n = e^{\mathrm{j}\arg\left((\rho_n\mathbf{I}-\mathbf{M}_n)\mathbf{u}_n - \frac{1}{2}\mathbf{b}_n\right)}$

ρ_n is a hyperparameter such that $\rho_n \geq \lambda_{\max}(\mathbf{M}_n)$

Eq. (32): $\rho_n = \lambda_{\max}(\mathbf{M}_n) + z_n$

Let NN decide!

Core idea:

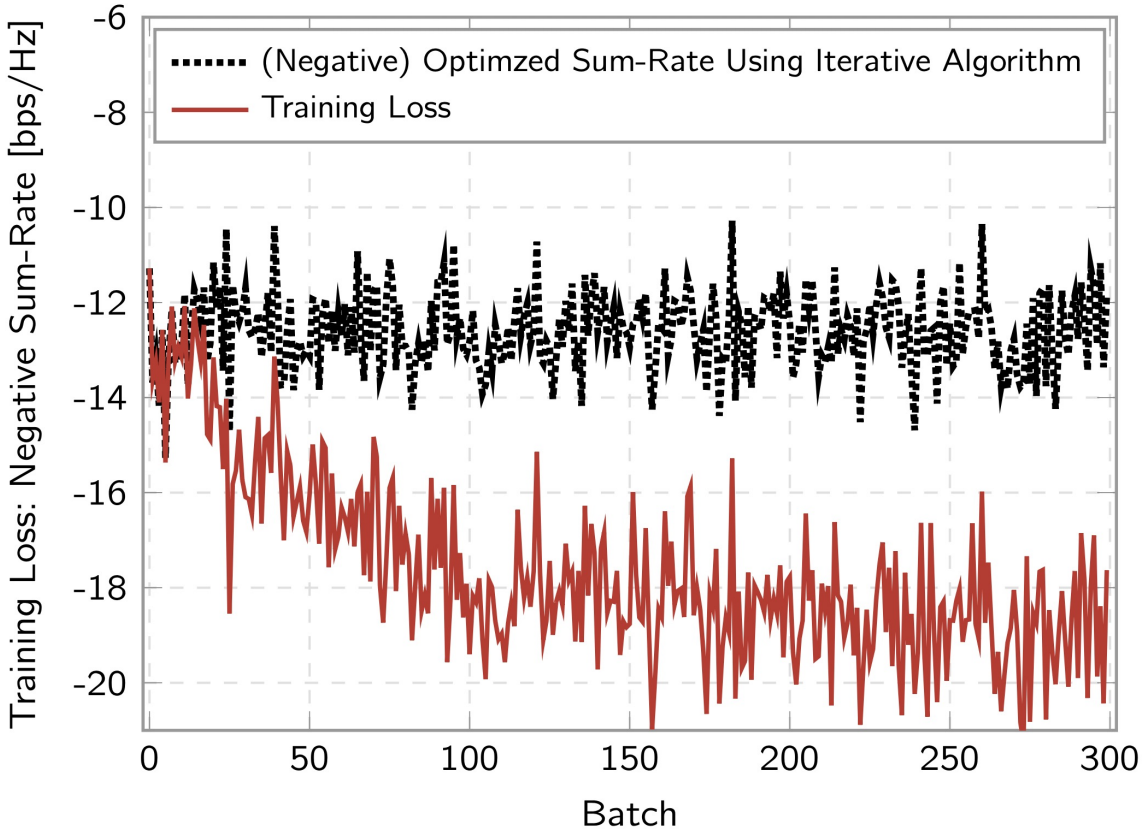
- Full convergence is impractical under limited latency.
- We allow **at most *L* iterations** and train a neural network to help achieve the best possible performance within this budget.

P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

Model Training

Module	Input MLP	GNN layers	Readout MLP
Ψ_a	$801 \to 406 \to 12$	$12 \to 12 \to 12$	$12 \to 10 \to 8$
Ψ_p	$801 \to 415 \to 30$	$30 \to 30 \to 30$	$30 \to 25 \to 20$

- DeepMIMO dataset [1].
- **Training data:** New York, Los Angeles, and Chicago (in total 28, 595 channel realizations).
- **Testing data:** Miami, Austin, Columbus, and Dallas.
- Varying user number $3 \leq K \leq 5$ [2].

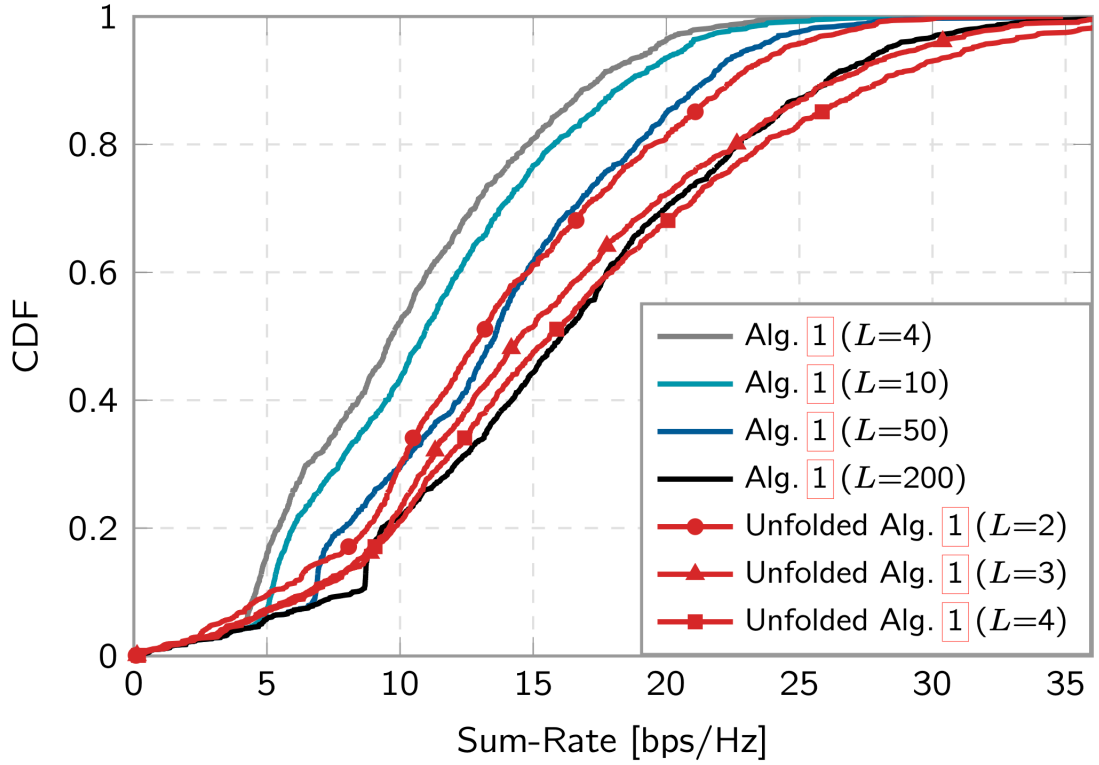


[1] A. Alkhateeb, “DeepMIMO: A generic deep learning dataset for millimeter wave and massive MIMO applications,” in Proc. of Information Theory and Applications Workshop (ITA), San Diego, CA, Feb 2019, pp. 1–8.

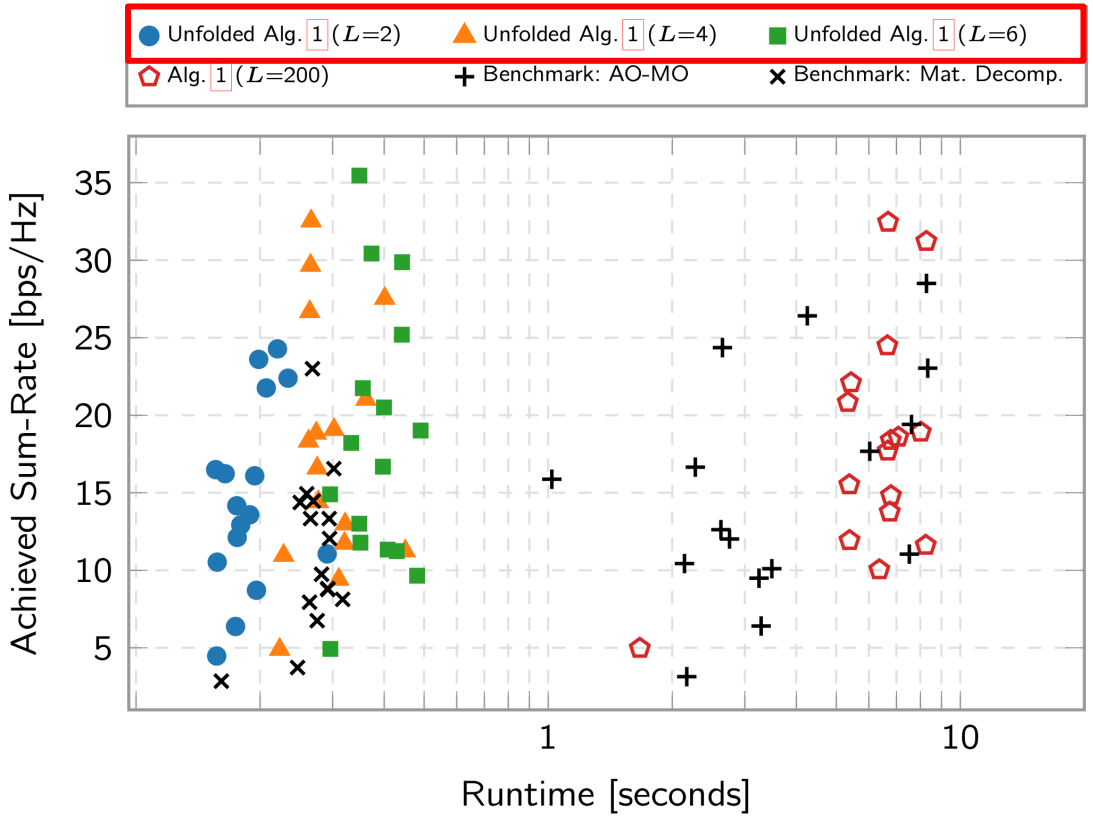
[2] P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

Performance Evaluation

4 iterations are already enough!



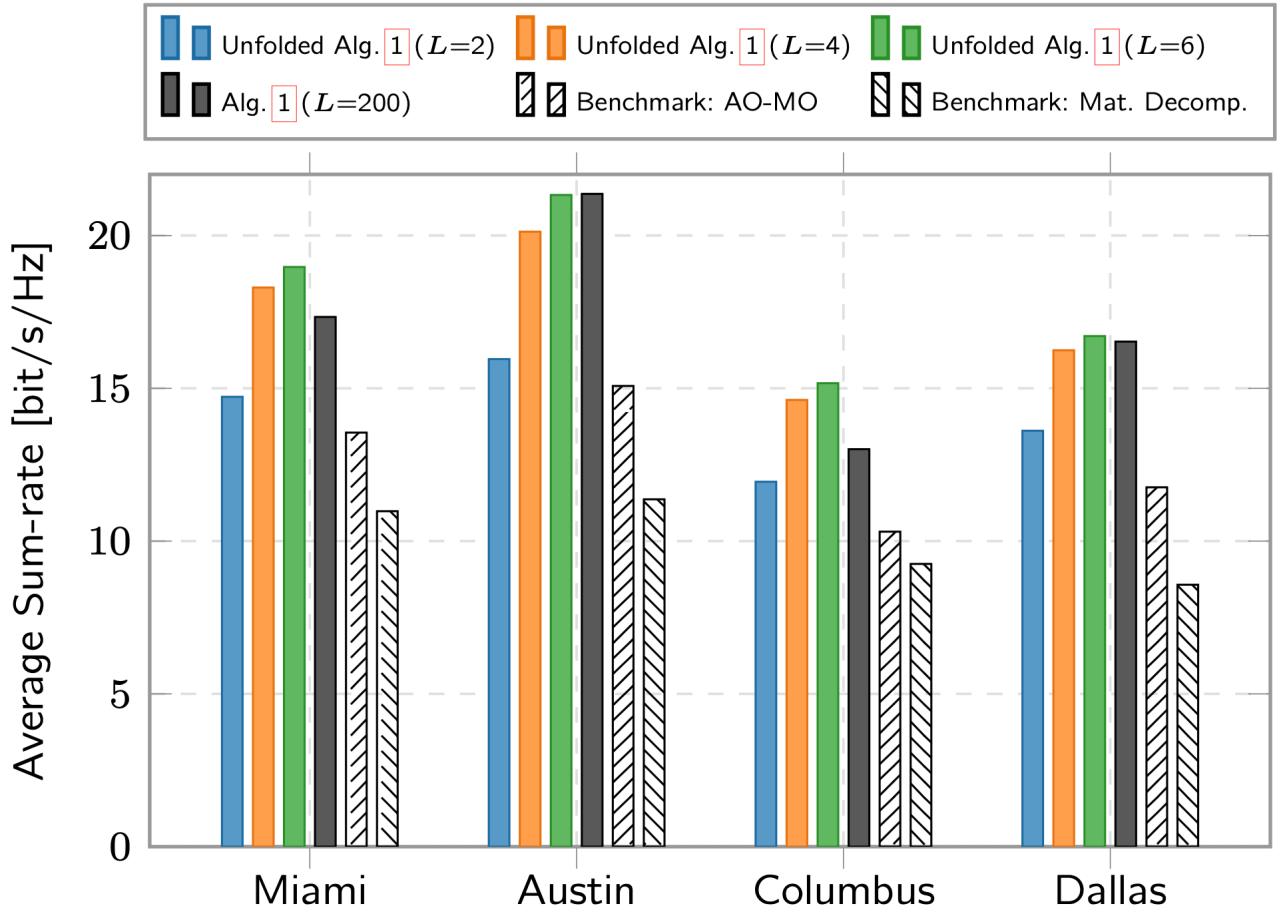
Enjoy both **high sum-rate** & **short runtime**!



P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

Performance Evaluation

Strong generalization across various environments



P. Zheng, M. J. Hossain, and A. Chaaban, "Fast Tri-Hybrid Beamforming via Deep Unfolding," in IEEE Transactions on Signal Processing, 2026, early access.

More to Explore

- Channel estimation
- Sensing and ISAC
- Signaling and feedback protocols
- More advanced data-driven beam management
- More advanced antennas: pattern, frequency, polarization, metasurfaces…
- Hardware fabrication and experimental validation

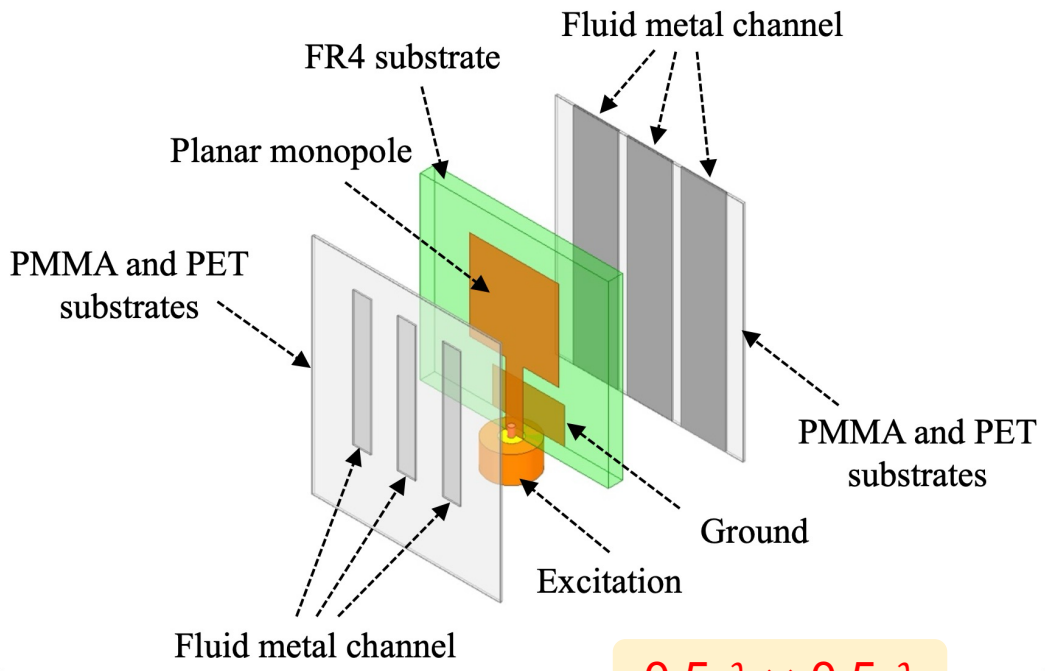
…

Outline

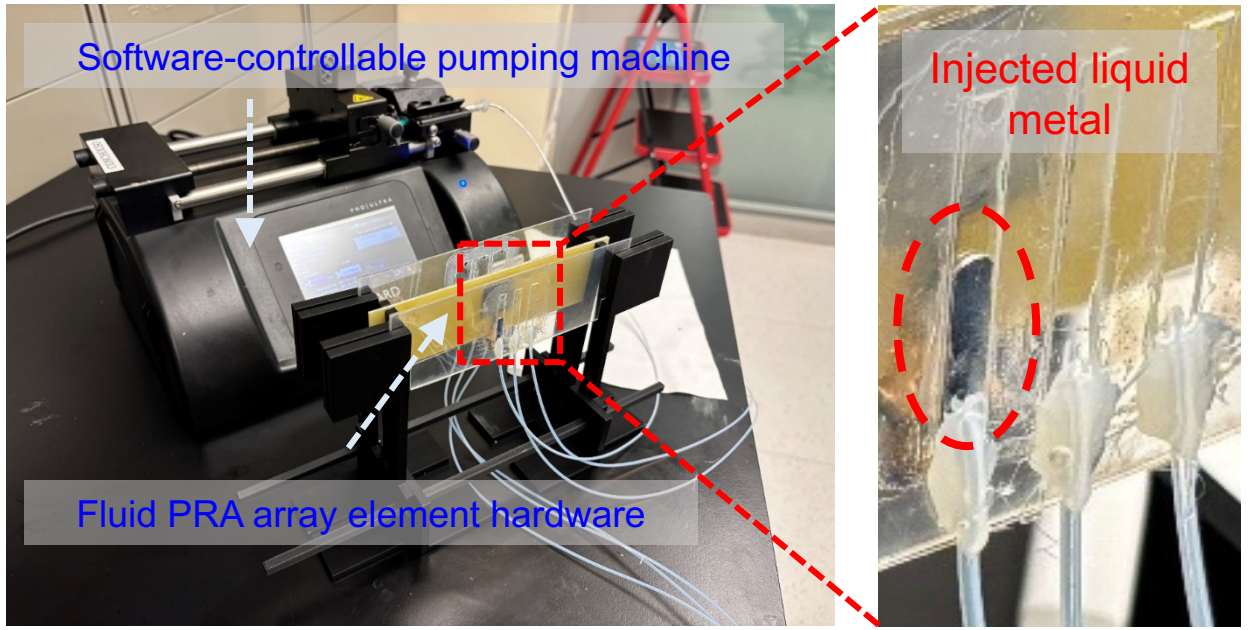
1. Background
2. Antenna Coding
3. Signal Processing for Reconfigurable Antennas
4. Reconfigurable Antennas for ISAC Applications
5. Reconfigurable Antennas for Tri-Hybrid MIMO
6. **Prototyping and Experiments**
7. Conclusions

A Prototyping Study: EM Reconfigurable Fluid Antennas

- Each fluid metal channel can be filled and emptied using liquid metal and a pumping machine.
- The radiation pattern of the unit can be changed by the parasitic coupling effect.



$0.5\,\lambda \times 0.5\,\lambda$

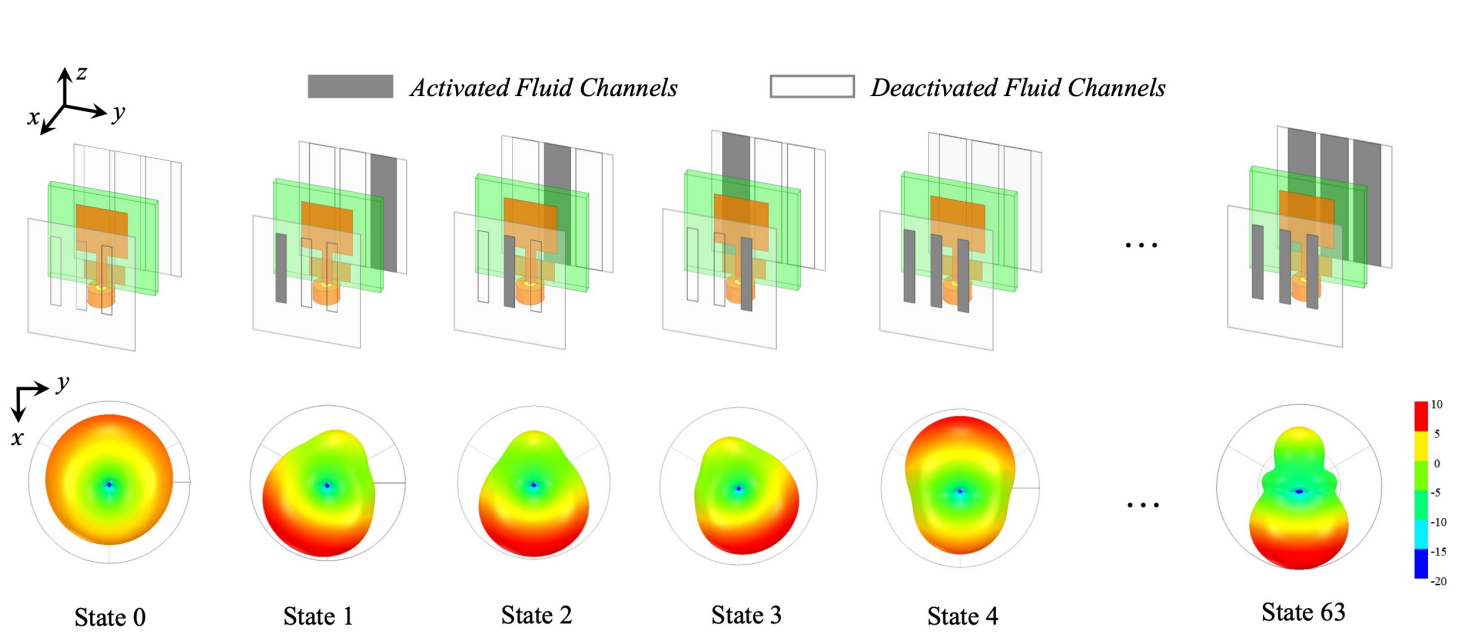


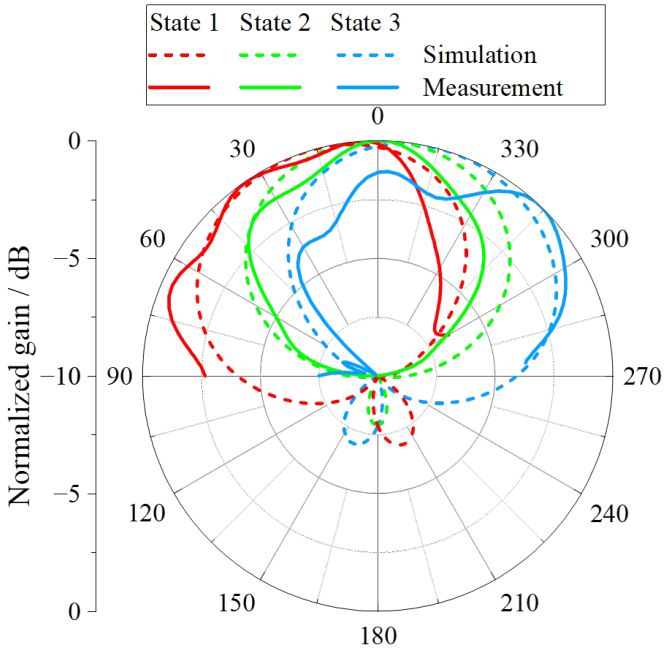
Galinstan liquid metal alloy: 68.5% gallium (Ga), 21.5% indium (In), and 10% tin (Sn)

R. Wang, P. Zheng et al., "Electromagnetically Reconfigurable Fluid Antenna System for Wireless Communications: Design, Modeling, Algorithm, Fabrication, and Experiment," in IEEE Journal on Selected Areas in Communications, vol. 44, pp. 1464-1479, 2026.

A Prototyping Study: EM Reconfigurable Fluid Antennas

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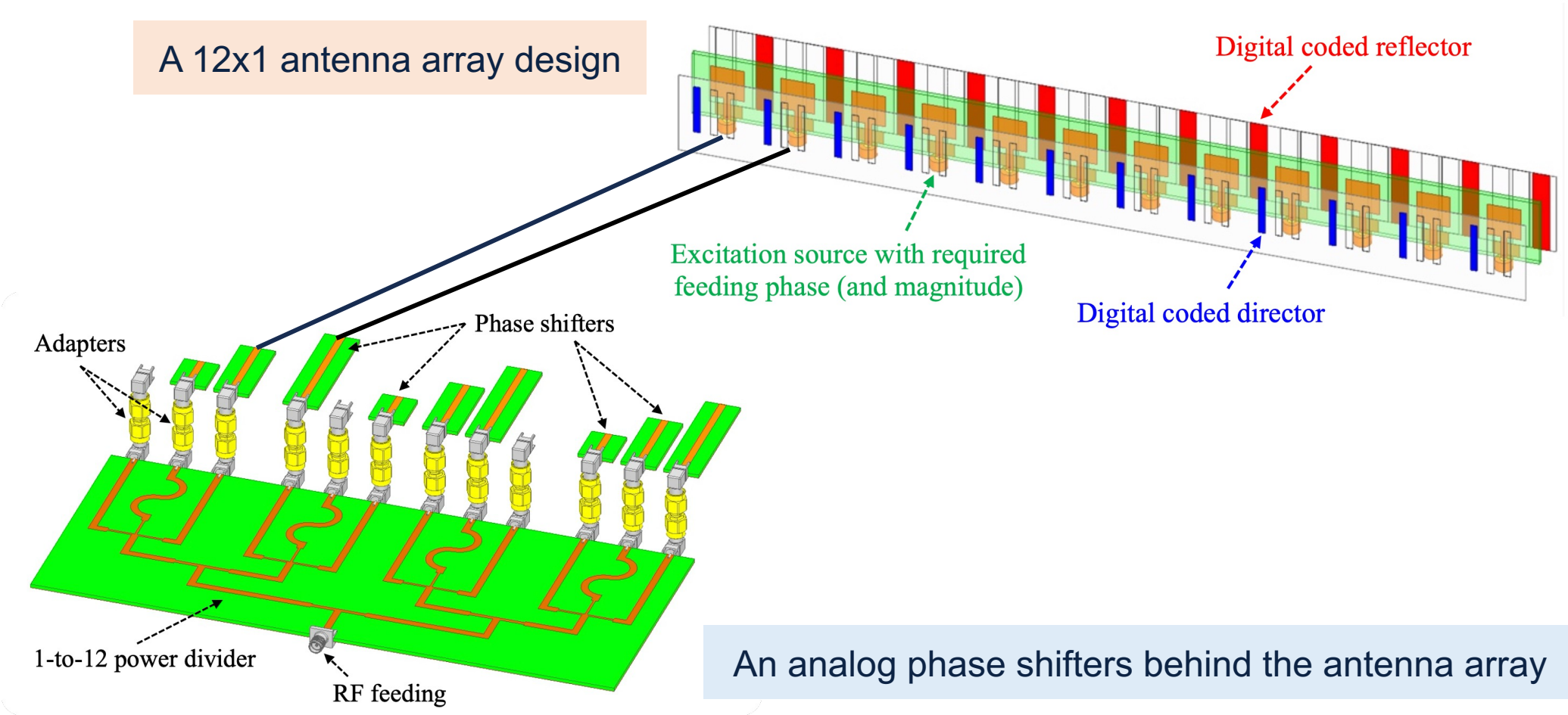




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A Prototyping Study: EM Reconfigurable Fluid Antennas

A 12x1 antenna array design

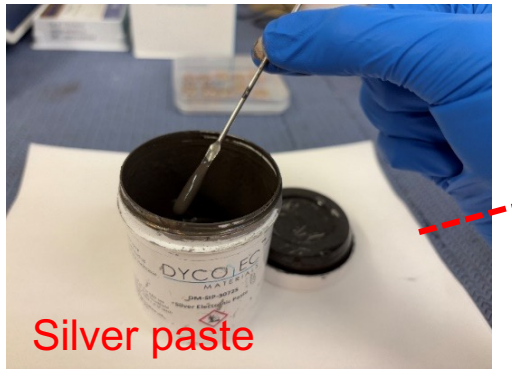


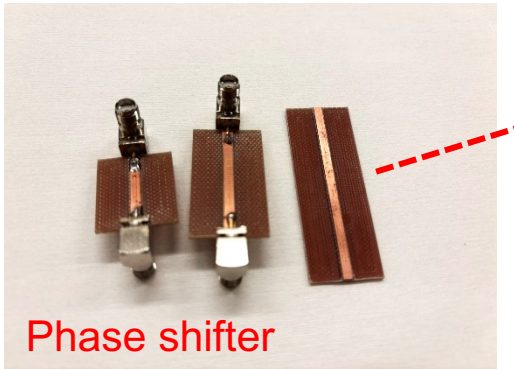
An analog phase shifters behind the antenna array

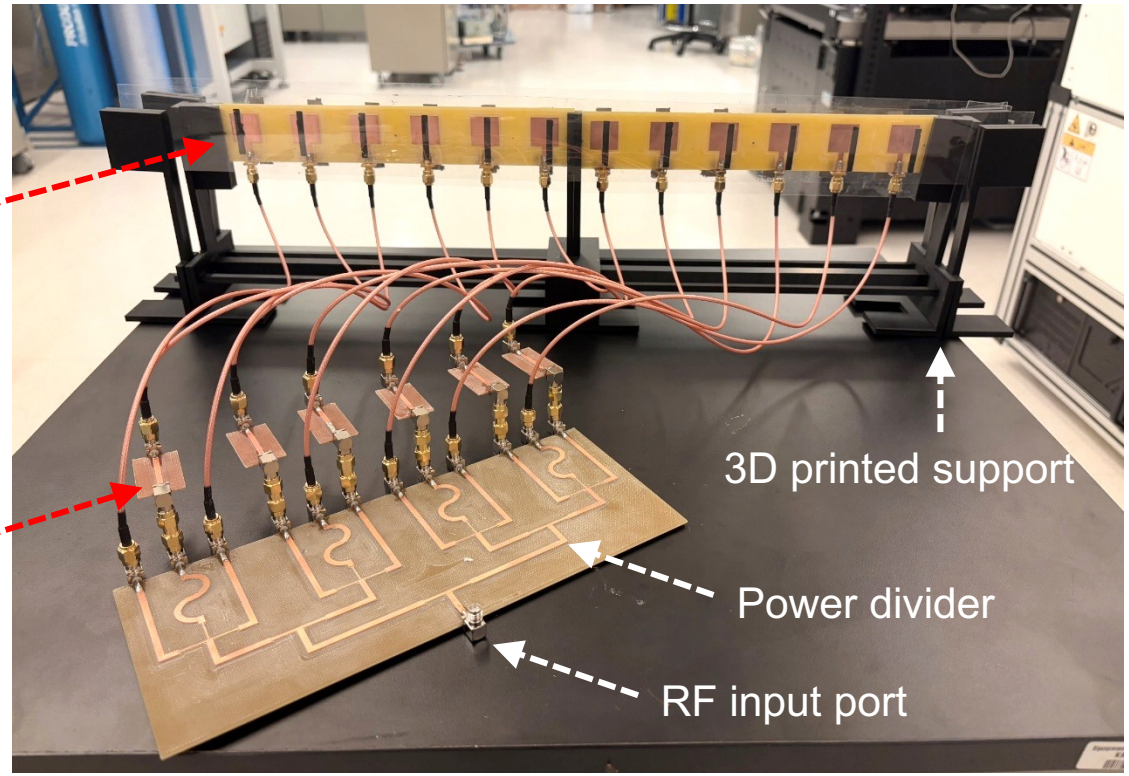
R. Wang, P. Zheng et al., "Electromagnetically Reconfigurable Fluid Antenna System for Wireless Communications: Design, Modeling, Algorithm, Fabrication, and Experiment," in IEEE Journal on Selected Areas in Communications, vol. 44, pp. 1464-1479, 2026.

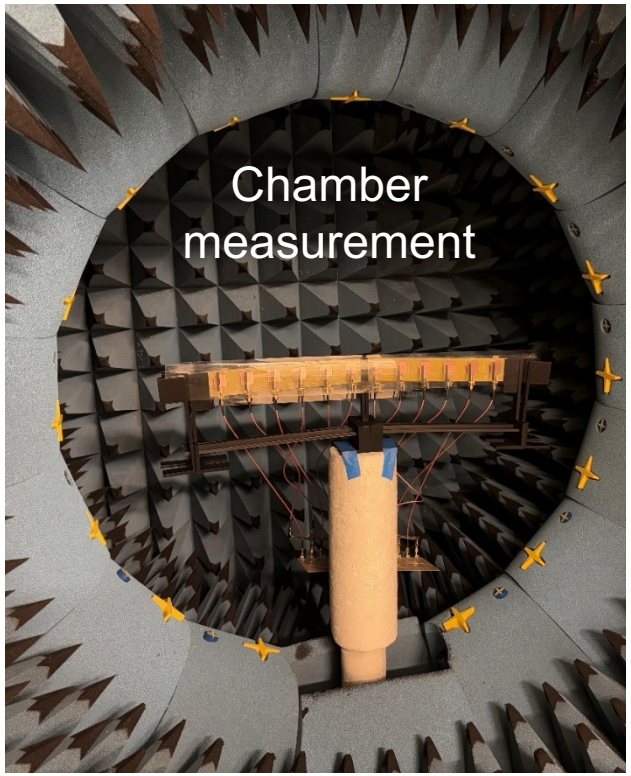
A Prototyping Study: EM Reconfigurable Fluid Antennas

The fabricated prototype at KAUST (3.55 GHz)



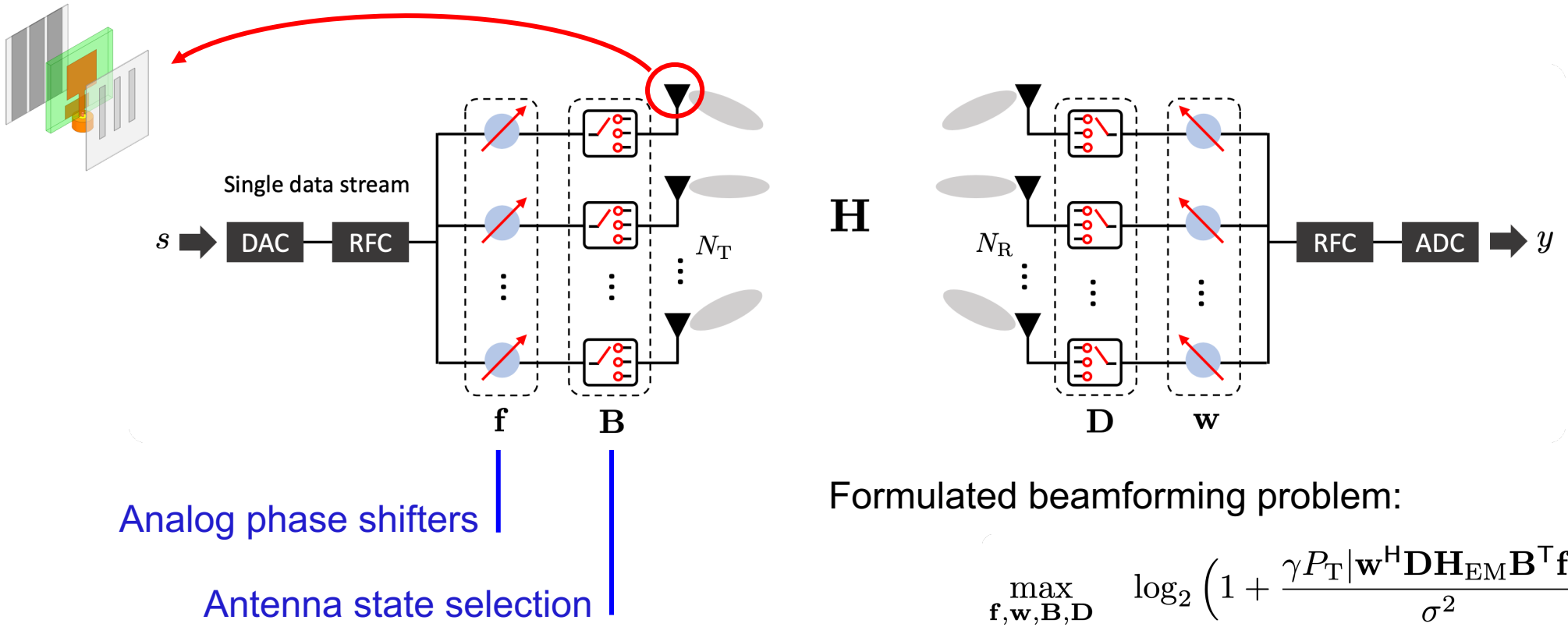






R. Wang, P. Zheng et al., "Electromagnetically Reconfigurable Fluid Antenna System for Wireless Communications: Design, Modeling, Algorithm, Fabrication, and Experiment," in IEEE Journal on Selected Areas in Communications, vol. 44, pp. 1464-1479, 2026.

A Prototyping Study: EM Reconfigurable Fluid Antennas



Formulated beamforming problem:

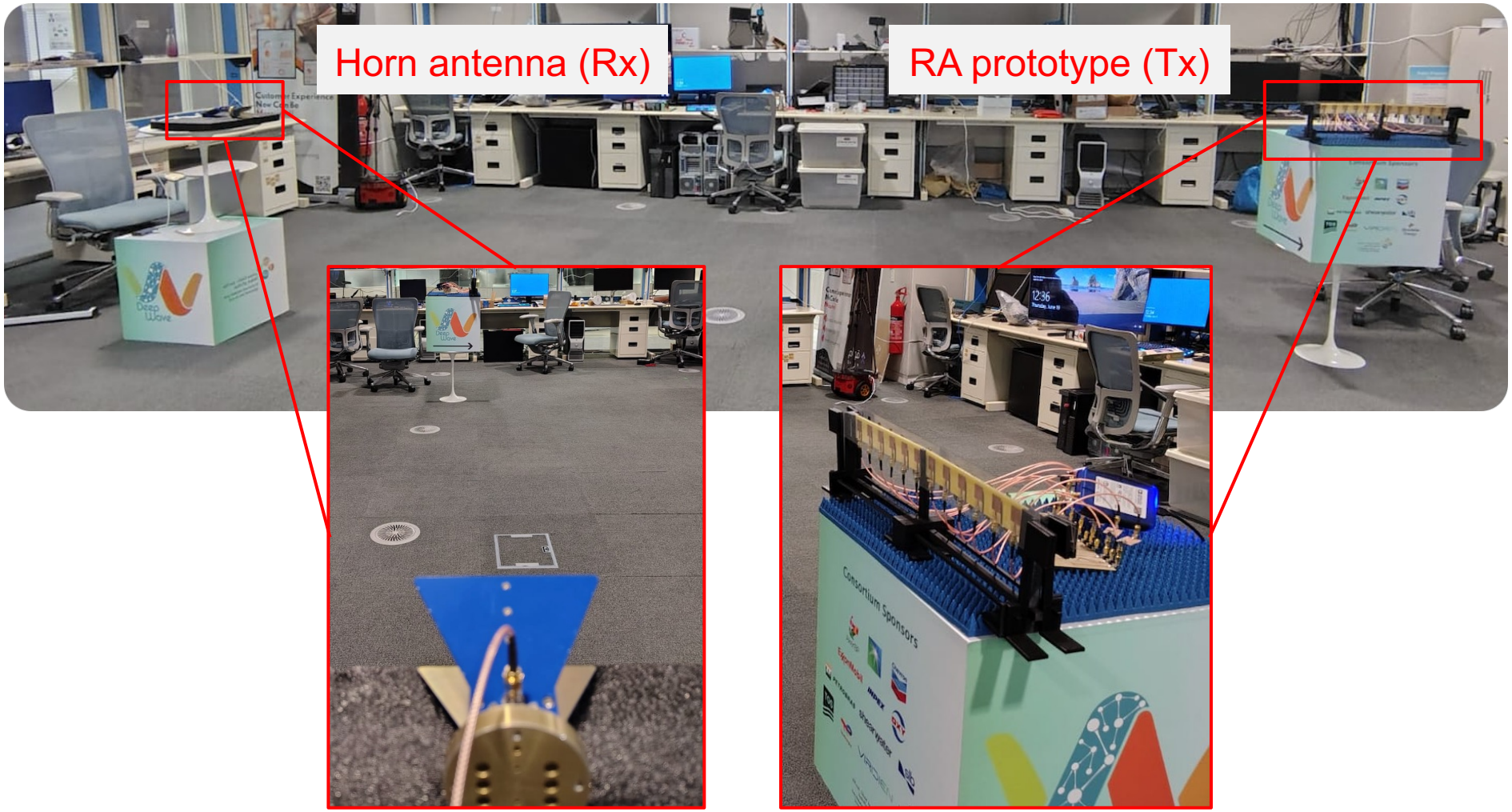
$$\max_{\mathbf{f},\mathbf{w},\mathbf{B},\mathbf{D}} \quad \log_2\left(1+\frac{\gamma P_{\mathrm{T}}|\mathbf{w}^{\mathsf{H}}\mathbf{D}\mathbf{H}_{\mathrm{EM}}\mathbf{B}^{\mathsf{T}}\mathbf{f}|^2}{\sigma^2}\right),$$

$$\text{s.t.} \quad |f_i|^2=\frac{1}{N_{\mathrm{T}}},\ |w_j|^2=\frac{1}{N_{\mathrm{R}}},\ (7),\ (9),\ (10),$$

R. Wang, P. Zheng et al., "Electromagnetically Reconfigurable Fluid Antenna System for Wireless Communications: Design, Modeling, Algorithm, Fabrication, and Experiment," in IEEE Journal on Selected Areas in Communications, vol. 44, pp. 1464-1479, 2026.

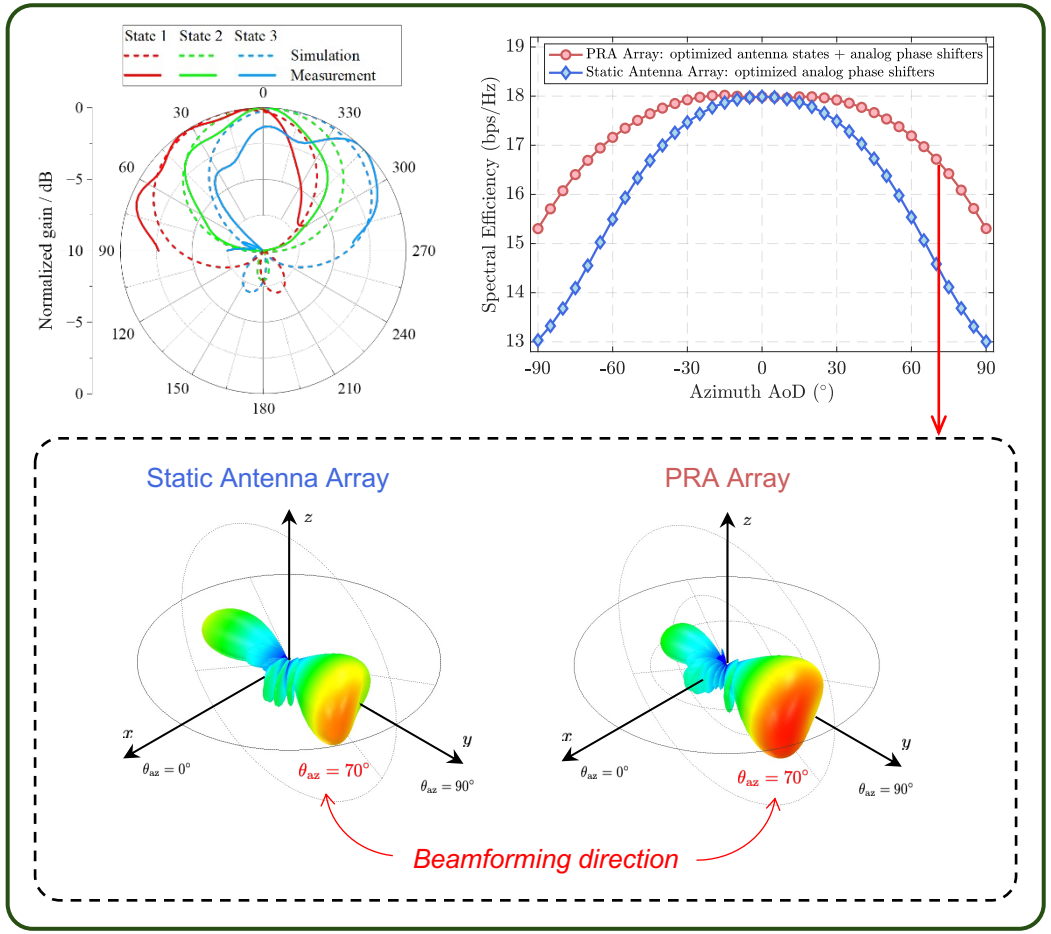
A Prototyping Study: EM Reconfigurable Fluid Antennas

Measurement setup (3.55 GHz, 4-QAM)

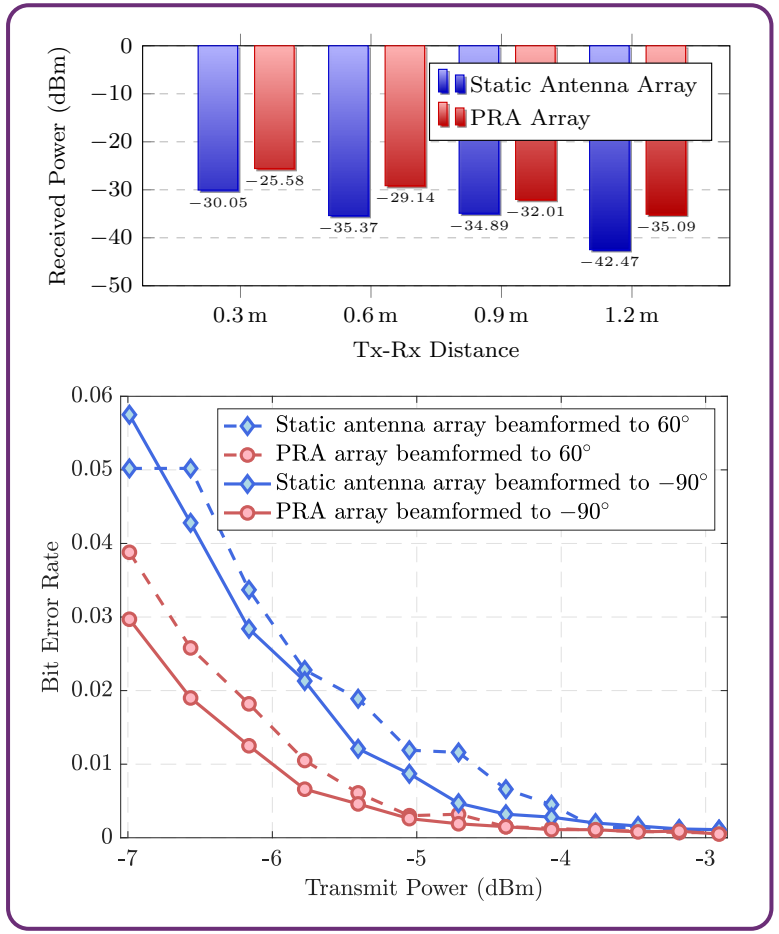


A Prototyping Study: EM Reconfigurable Fluid Antennas

Full-Wave Simulation: higher beamforming gain



Indoor Experiment: lower bit error rate



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A Prototyping Study: EM Reconfigurable Fluid Antennas

Key Insights

- RA array can provide a performance gain, using energy more efficiently and adaptively.
- Sophisticated signal processing is essential to unleash its potential.
- Slow reconfigurability is a major drawback of this design.
- The gain is not associated with liquid metal implementation, but the antenna pattern reconfigurability.
- More complete systems, more realistic testing environments, and more advanced antenna designs are still needed.

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